

Chapter 2

Prosody and Recursion in Coordinate Structures

Abstract

Based on evidence from the grammar of coordinate structures in English, this chapter proposes a compositional theory of prosody. A prosodic representation is assigned to complex expressions by recursively combining the prosodies of its parts, analogous to the compositionality assumed in semantics. A single mapping principle, recursively applied, captures an intricate set of metrical patterns. The system cyclically derives prosodic representations, integrating insights from the semantics, combinatorics, and syntax of coordinate structures. It is essentially an implementation of the ‘transformational cycle’ proposed in SPE. The new proposal, however, attempts to capture not only prominence relations—as the algorithm in SPE did—but also prosodic grouping and the strength of prosodic boundaries.

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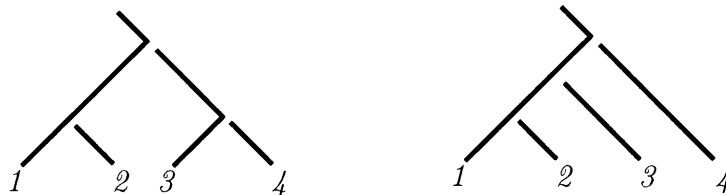
(1998, 29), is that compositionality is local. The denotation of any node in the syntactic structure is determined by combining the denotations of its daughters.²

The truth conditions computed by this compositional system tell us that stating (2a) is true and (2b) is false, although they consist of the same string of words. The difference in meaning is due to the difference in syntactic structure:³

(3) Syntactic Bracketing

a. $(p \wedge q) \vee (p \wedge r)$

b. $((p \wedge q) \vee p) \wedge r$



Along with assigning a meaning, the grammar also assigns a prosodic structure to its outputs. Just as the composition of the meaning of an expression reflects the recursive internal structure of the linguistic object, so does the prosody. Prosody disambiguates spoken renditions of (2a) and (2b).

Prosodic bracketing is reflected among other things by lengthening effects. These lengthening effects in coordinate structures have been investigated instrumentally e.g. in Lehiste (1973). Similar bracketing asymmetries were also used in the study on algebraic formulas in Lehiste et al. (1976). Prosodically induced lengthening effects were also investigated in Cooper and Paccia-Cooper (1980) and many subsequent studies, which are discussed in detail in chapter 4. Lengthening effects are illustrated in the following oscillograms. Consider in particular the length of ‘q’ and the length

²This strict locality assumption is called ‘context-free semantics’ in Dowty (2003). The treatment of variable binding using assignment functions in Heim and Kratzer (1998) departs from strict locality. Jacobson (1999) proposes a notion of ‘direct compositionality’ that adheres to strict locality.

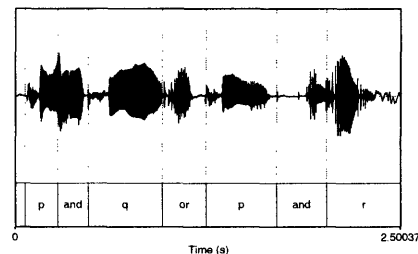
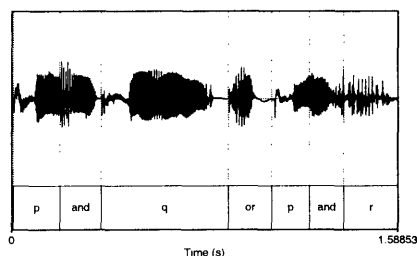
³The expression $((p \text{ and } q) \text{ or } p)$ seems redundant since logically it is equivalent to ‘p’ alone. But in natural language, disjuncts can be interpreted exhaustively, so the expression can mean $((p \text{ and } q) \text{ or just } p)$. This possibility of an exhaustive reading of the disjuncts is, I believe, responsible for the fact that disjunction can have an exclusive and an inclusive reading, which was shown by (Hurford, 1974). While I cannot motivate this claim at this point, I will stick with the example simply for the sake of illustration.

of the second ‘p’:

(4) Boundary Strength and Lengthening Effects

a. $(p \wedge q) \vee (p \wedge r)$

b. $((p \wedge q) \vee p) \wedge r$



The lengthening reflects the prosodic grouping of the expression, which in turn reflects the syntactic/semantic grouping. One key difference between the two structures is that in (2a), the boundary after ‘q’ is intuitively stronger than the boundary after the second ‘p’; the converse is true for (2b), where the boundary after ‘q’ is intuitively weaker than that after the second ‘p’.

I will indicate prosodic boundaries with the pipe symbol ‘|’. The strength of a boundary can be conceived of as a rank of the boundary on a discrete strength scale of boundary strengths. The boundary rank is encoded in the number of pipes (‘|’, ‘||’, ‘|||’, ‘|||’...). The boundary ranks reported in this chapter are based on native speakers’ intuitions. For the basic coordination cases they were also experimentally tested, as reported in chapter 5. The two structures above can then be represented as follows:⁴

(5) a. $|| p | \wedge q || \vee p | \wedge r ||$

b. $||| p | \wedge q || \vee p ||| \wedge r |||$

The strength of the boundaries directly reflect the syntactic constituency. Adjacent material separated from each other by weak boundaries and separated from the surroundings by stronger boundaries form constituent.

A similar notation is in fact used in Chomsky et al. (1957), where boundary symbols are hierarchically organized, and the organization is reflected by an index that

⁴In this chapter, I will not address the strength of the boundary between the connector and the following conjunct. This will be discussed in chapter 5.

reflects boundary rank. The representation for our example would be, where lower numbers reflect stronger boundaries:

(6) Syntactic Structure in Chomsky et al. (1957)

a. $\#_o p \#_2 \wedge q \#_1 \vee p \#_2 \wedge r \#_o$

b. $\#_o p \#_3 \wedge q \#_2 \vee p \#_1 \wedge r \#_o$

The phonetic lengthening effects illustrated in (4) can be made sense of on the assumption that they are a reflection of the boundaries in (5) and that stronger boundaries induce a higher degree of lengthening of the constituent preceding it compared to weaker boundaries.

The boundary ranks reported here reflect that the conjuncts/disjuncts connected by ‘and’ or ‘or’ are prosodically on a par, i.e. the perceived boundaries between the conjuncts/disjuncts are of equal strength. I will take this as evidence they form prosodic constituents of the same type. They will be presented as forming feet at the same level in metrical grid. A prosodic domain that contains several units that are prosodically ‘on a par’ in this sense will be henceforth called ‘prosodically flat’.

In (5a), there are two main disjuncts, and ‘p and q’ is prosodically on a par with ‘p and r’. Within each disjunct, there are two conjuncts separated by weaker boundaries, which again are on a par with respect to each other. In (5b), there are two main conjuncts, and the first, ‘p and q or p’, is prosodically on a par with the second, ‘and r’.

An obvious question is whether or not the boundaries have to be represented somehow in the prosodic structure or not. If the boundaries simply reflect syntactic branching it would seem redundant to code them in a metrical representation. A different way of looking at metrical representations however is that they are a *different way of representing syntactic information*—a way that is legible at the phonological interface. Furthermore, we will see evidence that prosody does not just encode tree structure, it also encodes information about the *derivational history* of a syntactic derivation, and the metrical representation will reflect information about *syntax* and *time*.

The idea that I want to pursue is that the overall prosody of an expression is computed based on decisions made locally, just as the overall denotation of an expression is computed by a stepwise combination of the denotations of its parts. Prosodic compositionality goes back to the transformational cycle of early generative approaches to phonology (Chomsky et al., 1957, Chomsky and Halle, 1968).

According to prosodic compositionality, the prosodic representation of (7a) is a proper subpart of the prosodic representation of (7b):

- (7) a. (p and q) or p
 b. ((p and q) or p) and r

This is not to say that the substring underlined in (7b) is *phonetically* identical to (7a). The two structures are identical at a more abstract level, just as the [k]s in spoken renditions of ‘cup’ and ‘cat’ are not phonetically identical, but are usually assumed to share an identical piece of information in their representation, the featural representation of [k]. Surface phonetic differences in the realization come about as a result of how phonological structure is implemented.

The main concern in this chapter is how the grammar assigns these relative boundary ranks, and prosodic structure more generally. The implementation of prosodic structure is discussed in chapter 4; experimental evidence is reported in chapter 5.

2.1.2 Outline

This chapter looks at the fragment of English consisting only of proper names and the functors ‘and’ and ‘or’. This fragment provides a simple mini-grammar in which we can establish some of the basics of the mapping from syntax to prosody. The argument proceeds in three steps.

What is the Generalization about Prosodic Boundaries?

First, we need to identify the generalization about the circumstances under which coordinate structures receive a ‘flat’ prosody, in which each conjunct is prosodically

on a par, and under what circumstances they are prosodically more hierarchically organized. Consider the following three structures:

- | | | | |
|-----|----|-----------------------|---|
| (8) | a. | $A \vee B \vee C$ | ‘Flat’ Prosody: $A \mid \vee B \mid \vee C$ |
| | b. | $A \vee (B \wedge C)$ | ‘Articulated’ Prosody: $A \parallel \vee B \mid \wedge C$ |
| | c. | $(A \vee B) \wedge C$ | ‘Articulated’ Prosody: $A \mid \vee B \parallel \wedge C$ |

Why does (8a) have a flat prosody, but (8b,c) have articulated prosodies? The proposal advanced in section (2.2) is that the reason is the same as in logical notation: (9a), but not (9b,c) would count as a well-formed and complete formula without parentheses.

- | | | | |
|-----|----|---------------------|------------------------|
| (9) | a. | $A \vee B \vee C$ | Associative |
| | b. | $A \vee B \wedge C$ | <i>Not</i> Associative |

Domains that have a flat prosody are domains in which the associative law holds. This is the case in (8a). When the associative law does not hold, then prosody reflects the syntactic bracketing, as in (8b,c). In those cases, constituency is reflected by boundary strength.

The prosodic generalization can be stated in terms of ‘levels of embedding’. Levels of embedding are domains that are associative.⁵

(10) ASSOCIATIVE DOMAINS

Levels of embedding (to be called ‘Cycles’) are associative domains.

Levels of embedding can be nested. Operators within lower levels of embedding take narrow scope relative to operators in higher levels of embedding, since they are syntactically more embedded in the structure. The notion of ‘level of embedding’ lets us state the generalization about prosody:

(11) SCOPALLY DETERMINED BOUNDARY RANK:

Iff Boundary Rank at a given level of embedding is n , the rank of the boundaries between constituents of the next higher level is $n+1$.

⁵A level of embedding with two elements trivially satisfies the associative law since no rebracketing is possible.

Constituents within a single level of embedding are set off by prosodic boundaries of equal rank; constituents within a lower level of embedding are set off by weaker boundaries than constituents within a higher level of embedding.

This hypothesis makes predictions for the prosody of coordinate structures. These will be discussed in detail in this chapter. The generalization directly relates prosody and scope, and captures the generalizations about prosody in coordinate structures.

What is the Syntax of Coordinate Structures?

What is the syntax of coordinate structures? This question is addressed in section 2.3. I will present evidence from combinatorics and from c-command tests that a binary-branching syntax combined with a cyclic view of syntactic derivations has just the right properties: It provides the correct combinatorial power and allows us to maintain the standard notion of c-command.

The cyclic approach to syntax exploits the property of recursiveness of language to define steps in the derivation. Grammar assembles structures in a workspace. Upon completion, a workspace is ‘spelled out’, which means that it is phonologically and semantically interpreted. These work-spaces form processing units in the derivation, and are called ‘cycles’. The outputs of the cycles are the outputs of the grammar.

Any output of the grammar, i.e. the output of any cycle, can re-enter a new cycle as an atom, contributing the semantic and prosodic interpretation assigned at the earlier cycle. This is simply the generally assumed recursiveness of language, and has entered recent minimalist theorizing under the keyword ‘multiple spell-out’ (Uriagareka, 1999). The key for any cyclic theory of syntax is to specify which domains constitute a cycle. This question is at the heart of much of the current research in the minimalist theorizing within the framework of ‘Phase Theory’, following Chomsky (2001).

The new proposal here is that cycles correspond to the ‘levels of embedding’ defined above: they are associative domains. In other words, whenever three elements are combined in such a way that the associative law does not hold, then they must be assembled in two separate cycles. Only associative domains are built in a single

cycle.

(12) Generalization about Syntactic Cycles

Each Syntactic Cycle is an Associative Domain.

It is the interface to semantics then that determines which nodes are cyclic nodes in syntax and which nodes are not. Nodes become cycles and are sent off to be interpreted in semantics and phonology in order to obtain a certain semantic interpretation. This chapter focuses on coordinate structures, but the approach to cycles in terms of associativity will be extended to other domains in chapter 3. We will see in chapter 6 that there is at least one additional factor that can render a syntactic node into a cyclic node. This will be a case where the interface to phonology is at play.

Evidence standard tests for c-command discussed in section 2.3 suggests that for associative domains that are assembled in a single cycle, grammar imposes right-branching structures. This, I claim, is due to the fact that within a single cycle, grammar generally can only construct right-branching structures. The following principle captures this property of grammar, which I assume to be a language-universal:

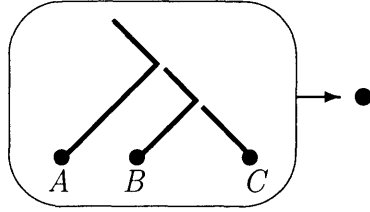
(13) RIGHT-BRANCHING CONJECTURE

Each cycle consists of a right-branching structure.

This is an adapted version of the principle proposed in Haider (1993, 2000), who proposed that ‘shell-structures’ are always right-branching. A similar principle was proposed in Phillips (1996, 19), who presents evidence both from syntactic constituency tests and from parsing preferences observed in the psycho-linguistic literature. The system proposed here shares with Phillips’ system the property that structures in language are right-branching unless interpretation motivates a left-branching structure.

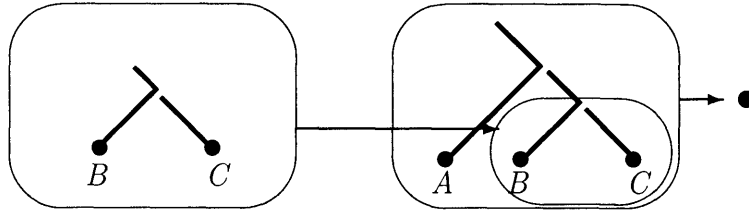
The structures in (8) can be built in one cycle. In the illustrations, each oval constitutes a cycle (also called ‘workspace’). A cycle counts as an ‘atom’ for any higher cycle, and I sometimes represent them by filled circles to simplify the representation.

(14) (8a), Derived in One Cycle

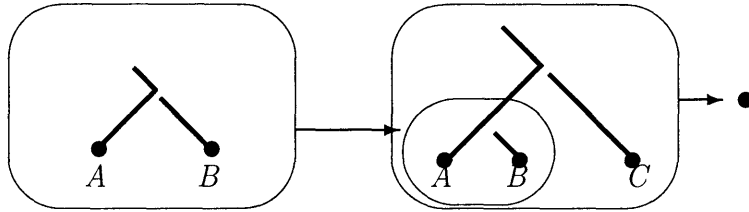


The structures (8b,c), on the other hand, have to be built in two cycles. Although (8a,b) are both right-branching, they differ in derivational history, and (b) consists of two separate cycles.⁶

(15) a. A or (B and C) (8b): Derived in Two Cycles



b. (A or B) and C (8c): Derived in Two Cycles



The cyclic theory based on binary branching has the right combinatorial properties and captures generalizations about c-command.

⁶Dresher (1994) observes that both types of right-branching structures are treated in the same way in the accent annotations in the Masoretic Bible. He convincingly argues that the accents reflect prosodic structure, and they faithfully distinguish right-branching from left-branching. The very notational system used makes it impossible to distinguish the two different right-branching structures. There are two possibilities: (i) the transcribers were forced by the notation they used to ignore the distinction, although it is present in the prosody; or (ii), the distinction is not prosodically realized, and the accent system faithfully transcribes the how the text is chanted. The latter possibility would be a problem for the present approach. It would imply either that Tiberian Hebrew is different in how it reflects prosody, or that the theory presented here is incorrect.

How does the Syntactic Cycle Interact with Phonology?

The final part of the argument shows how the syntactic derivation can be used to derive the correct prosodic structure. The correct prosody can be derived on the assumption that the phonological cycle applies upon completion of a syntactic cycle, following Bresnan (1971) and more recently Chomsky (2001). Once completed, the structure assembled within a cycle is ‘spelled-out’ and is mapped to a single prosodic foot in the metrical grid.

(16) PROSODIC MATCHING

a. Concatenate

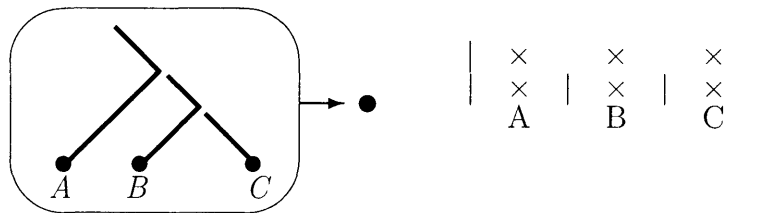
Concatenate the prosodic representation of the elements in the domain aligning their top lines and filling the columns where necessary.

b. Project

Create a new top-line grid line n by projecting all grid marks on line $n-1$, and mapping them into a single foot on line n .

‘Concatenate’ assures that constituents that are combined in a cycle start out on an equal footing in their prosodic representation. ‘Project’ derives a ‘flat’ prosody for elements assembled in a single cycle:

(17) A or B or C (8a): Derived in One Cycle



The pipes in the prosodic representation mark foot boundaries in the prosodic notation used here, the metrical grid. The height of columns of pipes is interpreted as the boundary rank of a juncture. The structure in (17) is ‘flat’ in that all elements are separated by boundaries of the same rank (here: a column of height 1). The metrical grid was proposed in Liberman (1975) to encode information about the timing of an

utterance, and Selkirk (1984) explicitly relates the grid to phonetic correlates such as lengthening and pauses. I will discuss in detail how the representation is to be interpreted and how it relates to alternative representations in section 2.4.

When elements are combined in such a way that the associative law does *not* hold, then they have to be assembled in separate cycles. The output of a first cycle enters a new cycle as an atomic element. Since after each cycle spell-out maps the output to a prosodic foot in new grid line, the prosody reflects the way the structures are put together by syntax, deriving the following grids for (8b,c) respectively:

(18) Metrical Grid

a. A or (B and C) (8a)

	×		×		×	
	×		×		×	
	A		B		C	

b. (A or B) and C (8b)

	×		×		×	
	×		×		×	
	A		B		C	

This accounts for the fact that a more articulate prosody is derived for building (8b,c), but not for (8a). The crucial difference between the structures is that only (8a) is an associative domain and is therefore assembled in a single cycle.

While there are only two different syntactic trees involved in the three structures in (8), there are three different syntactic derivations. The prosody keeps track of the derivational history, and can be seen as another way of representing syntactic information. In addition to the syntactic relations represented in the tree, prosody reflects how it was assembled: prosody includes information about time.

The theory of syntax—phonology mapping presented here exploits the recursive-ness of language to capture prosodic differences. Prosody reflects syntactic bracketing encoded in tree structures, but also the derivational history of how these tree structures were assembled. There are exactly as many prosodic structures as there are syntactic derivations, and the syntactic representation using binary branching trees allows us to maintain the standard notion of c-command in syntax. A single mapping principle (16) is sufficient to derive the intricate prosodic pattern of coordinate structures, and relates it directly to interpretative properties such as associativity

and scope.

The interaction between cycles and prosody has the effect that a prosodically articulated structure is derived under the same circumstances that orthographic brackets are placed in the notation of logical formula: when the associative law does not hold. The prediction is then that structures are prosodically ‘flat’ unless interpretation motivates a more articulate prosodic structure.

2.2 Boundaries and Associativity

This section presents evidence that prosodic structure encodes the different bracketings of coordinate structures by assigning boundaries of different strength. Grammar differentiates boundaries of different strength only in cases in which the associative law does not hold.

2.2.1 Inducing Bracketing

Utterances consisting only of proper names and connectors can occur as fragment answers to questions:

- (19) Who went to the forest?
- a. Lysander and Hermia.
 - b. Lysander or Hermia.

The fragment answer (19a) can be seen as a shorthand for one of the following expressions, which seem to be identical in their truth conditions. In fact, fragment answers are often analyzed as being derived from full sentences by ellipsis (Merchant, 2003).⁷

⁷Some authors (e.g. Chomsky (1957, 36) and Harris (1957, 318ff)) have argued that sentences such as (20a) are derived from such sentential coordinate structures as (20b) by coordination reduction, an operation that deletes identical information in the two conjuncts. The appeal of this approach is that it offers the prospect to reduce all instances of ‘and’ to a Boolean connector that relates propositions with truth values. The following example illustrates that coordination reduction cannot be the right theory though:

- (20) a. Lysander and Hermia went to the forest.
- b. Lysander went to the forest and Hermia went to the forest.

The precise bracketing between the conjunctions with more than two elements often does not seem to affect the truth conditions. The associative law holds for coordination between propositions. The following three answers seem to have the same truth conditions in the context given:⁸

- (21) Who went to the forest?
- a. Lysander | and Demetrius | and Hermia.
- b. Lysander | and Demetrius || and Hermia.
- c. Lysander || and Demetrius | and Hermia.

The most natural answer in this context would be (21a). The more articulated prosodies in (21b,c) seem gratuitous, unless there is something in the context that motivates grouping two of the three individuals together. Structures (21b,c) introduce a presupposition that requires a certain grouping to be salient. Thus, if the prosodically articulated answer in (21c) is used, the listener would surely assume that the speaker has some reason to group Demetrius and Hermia together, possibly because they went to the forest together, or because they were grouped together in the conversation before.

-
- (1) a. Lysander and Hermia gathered in the forest.
- b. *Lysander gathered in the forest and Hermia gathered in the forest.

This does not necessarily imply that coordination is not always Boolean coordination. Schein (1997) and Winter (2001) both propose theories that only use Boolean ‘and’, and yet they do not analyze coordinate structures always via coordination reduction.

⁸I will not be concerned here whether a coordination should be treated as the projection of the connector or a projection of the first conjunct as in Munn (1993). I will assume however, that coordination in general returns an element of the same semantic type, and thus can apply recursively to connect more than 2 elements.

Consider the following context, in which ‘Hermia’ is contrasted with ‘Demetrius and Helena’. This context favors (21c):⁹

(22) Lysander and Hermia?

No. Lysander || and Demetrius| and Helena.

The nature of the presupposition and how it affects prosody is discussed in chapter 7. In the following, I will look at cases where prosody affects truth conditions of expressions. Adverbs such as ‘together’ and ‘alone’ interact with prosodic grouping in ways that affect truth conditions. The answer in (23a) is compatible with Lysander going on his own, but (23b) cannot have this reading.¹⁰

(23) a. Lysander || and Demetrius | and Hermia together.

b. Lysander | and Demetrius || and Hermia together.

The adverb ‘both’ can also be used to induce bracketing (cf. Winter (2001)). ‘Both’ presupposes that there are two conjuncts, and not more. Now consider:¹¹

(24) a. Both Lysander || and Demetrius | and Hermia.

b. Both Lysander | and Demetrius || and Hermia.

The flat prosody is impossible with more than two conjuncts and ‘both’:

(25) * Both Lysander, | [and] Demetrius, | and Hermia.

⁹It also seems to be possible to use the flat prosody in the answer, and contrast ‘Lysander and Hermia’ with ‘Lysander, Demetrius, and Helena’. The point is that information structure *can* motivate prosodic bracketing.

¹⁰And the additional sub-grouping of Lysander and Demetrius requires that there is some motivation for grouping those two together to the exclusion of Hermia.

¹¹Chomsky (1965, 196) alternates ‘and’ with ‘as well as’ in order to illustrate embedded coordination:

(1) John, as well as Mary and her child.

‘as well as’. I am not sure whether ‘as well as’ necessarily requires an articulated prosody when combined with ‘and’ or not.

The interaction of ‘both’ with prosodic bracketing can be made sense of if (24a,b) involve a coordination distributing over two propositions, one of which includes a plural subject consisting of the two conjuncts that are grouped together more closely. This type of collective reading may also be involved when using adverbs such as ‘together’. Collective readings are analyzed in Winter (2001) as involving a covert collective operator that creates a plurality (but see Schwarzschild (1996) for an alternative view). Consider the following example (cf. Winter (2001, 31), also Schein (1993)):

- (26) a. Lysander and Demetrius weigh exactly 200 lbs.
 b. Lysander weighs exactly 200 lbs and Demetrius weighs exactly 200 lbs.

Sentence (26a) has a collective reading that (26b) lacks, which is that Lysander and Demetrius together weigh 200 lbs. Both sentences have a ‘distributive reading’, in which Lysander and Demetrius each weigh 200 lbs. In cases that involve a coordination with a collective interpretation the bracketing of coordinate structures can play a crucial role.

- (27) a. Lysander | [and] Demetrius | and Hermia weigh 200 lbs.
 b. Lysander | and Demetrius || and Hermia ||| weigh 200 lbs.
 c. Lysander || and Demetrius | and Hermia ||| weigh 200 lbs.

Each sentence has a distributive reading in which each of the three weighs 200 lbs. Example (27b) has an additional reading, where Lysander and Demetrius together weigh 200 lbs and Hermia does so alone. This collective reading that groups Lysander and Demetrius together is absent in (27c), which in turn has a reading in which Demetrius and Hermia together weigh 200 lbs and Lysander weighs this much alone.

Consider also the following example. The answer in (28a) seems inappropriate, while (28b) and (28c) differ with respect to who will have to share an apple:

- (28) Two apples were given out, but I don’t know to who. Who was given an apple?
 a. # Lysander | and Demetrius | and Hermia.

- b. Lysander | and Demetrius || and Hermia.
- c. Lysander || and Demetrius | and Hermia.

Hoeksema (1983) observes the following case, where again a collective reading is involved, licensing a certain reading of the reciprocal:

- (29) (Blücher and Wellington) and Napoleon fought against each other near Waterloo.

It is not necessary to make the subset of individuals into a constituent in order to form an antecedent for a reciprocal interpretation. The following sentence can have the same reading as (29):

- (30) The generals fought against each other.

But it is hard if not impossible to give the following sentence the reading intended in (29):

- (31) Blücher and (Wellington and Napoleon) fought against each other near Waterloo.

Another way to force the bracketing is to alternate ‘and’ and ‘or’. With alternating functors it seems that only the articulated two bracketings are possible.¹² This observation was also made in Min (1996):

- (32) a. ?? Lysander | and Helena | or Demetrius.
 b. Lysander | and Helena || or Demetrius.
 c. Lysander || and Helena | or Demetrius.

¹²It may be possible to pronounce at least (32) with a pretty flat prosody, and in fact several renditions of the recordings collected in the experiment reported on in chapter 5 sound like this happened. Maybe lower bracketings can be ‘washed out’, at least in fast speech. But a careful pronunciation would certainly involve either of the two articulated prosodies, depending on the intended meaning.

While in the earlier examples, it was some overt or covert scope-taking element that seemed to interact with the bracketing in coordination structures, in the example in (32) it is the scope of ‘and’ and ‘or’ with respect to each other that is at stake. It seems that in each case some covert or overt scope-taking element takes scope over some conjuncts but not others. The scope of sentential adverbs in coordinate structures also interacts with prosodic grouping:

- (33) Who went to the forest?
- a. Lysander || and probably | Demetrius || and Hermia.
 - b. Lysander ||| and probably || Demetrius | and Hermia.

In example (33a), only Demetrius’ going to the forest is uncertain, whereas in (33b) the whereabouts of both Demetrius and Hermia are not certain. When ‘probably’ takes scope over a single conjunct, then the conjuncts can be separated by boundaries of equal rank, as in (33a); but when ‘probably’ takes scope over the two last conjuncts, these two are grouped together and separated from the first conjunct with a stronger boundary (33b).¹³

Other sources for sub-groupings in conjunctions exist. Consider the difference in phrasing in the following examples (cf. Gleitman, 1965, 277):

- (34) What’s on the menu?
- a. Ham | and tongue | and tuna fish | and pastrami.
 - b. Ham | and eggs || and tuna fish || and pastrami.

Here food items that form single dishes are grouped together. Items can also be coordinated to create complex names, here names of linguistic publications (Bob Ladd, p.c.):

¹³Of course, it is possible for ‘probably’ to take scope over a single conjunct even when there is a stronger boundary after the first conjunct. The difference in prosody from (33) is then that the boundary following ‘probably’ is weaker than the boundary separating the two last conjuncts:

- (1) Lysander ||| and probably | Demetrius || and Hermia.

(35) What did you read?

Chomsky | and Halle || and Kiparsky | and Kiparsky.

What is the commonality among all cases in which prosodic boundaries of different ranks are induced? There is a simple formal property that all cases that show a more articulated prosody share, which is that they are not *associative*.

2.2.2 The Associative Law and Prosodic Domains

A constituent consisting of three elements x , y , and z is associative if the meaning¹⁴ do not change under either order of composition (The parentheses ‘()’ stand for functional application, as defined in Heim and Kratzer (1998)):

(36) ASSOCIATIVE LAW

$$([[x]])[[yz]] = [[xy]]([[z]])$$

Structures with a flat prosody are associative, in other words the bracketing does not carry any important information.¹⁵

Consider an analogy to the orthographic convention on placing parentheses in logical notation. Formulas (37a,b) are well-formed as they are, while formula (37c) seems incomplete:

- (37) a. $A \vee B \vee C$
b. $A \wedge B \wedge C$
c. $A \vee B \wedge C$

Parentheses are redundant in (37a,b), since in cases where ‘and’ and ‘or’, respectively, are iterated the associative law holds; but it does not hold when ‘and’ and ‘or’ alternate, as in (37c), hence the bracketing becomes crucial.

The generalization for the coordinate structures that emerged in the previous section is that in cases when $n > 2$ elements are combined, a prosodically articulated

¹⁴Relevant are at least the truth conditions and presuppositions, possibly also implicatures.

¹⁵In math, functions that are associative are sometimes called ‘flat’, e.g. ‘flat’ is the attribute used in *Mathematica* to mark a function as associative.

structure is obligatory exactly when the associative law *does not* hold, be it due to the presence of overt or covert additional scope-taking operators or because we alternated the functors ‘and’ and ‘or’, which take scope with respect to each other.

The correlation between ‘flat’ prosody and associativity, I want to claim, is not accidental. I propose the following hypothesis:

(38) THE ASSOCIATIVE-PROSODIC-DOMAIN-HYPOTHESIS

The associative law holds in the composition of constituents iff they are on a par prosodically.

The hypothesis is descriptively correct for the domain of evidence that we have looked at so far. It might seem that the solution is very specific to the domain under investigation: coordinate structures. In chapter 3 it is shown that it applies to other domains as well.

Let us call a sequence of elements, combined in such a way that the associative law holds, a single level of embedding, or a ‘cycle’.

(10) ASSOCIATIVE DOMAINS

Levels of embedding (to be called ‘Cycles’) are associative domains.

In cases where grammar combines more than two elements, it makes a decision whether to map them to a single level of embedding or whether to map them to two different levels of embedding.

Depending on the grammatical framework one assumes, the syntactic definition of level of embedding will have to be adjusted accordingly. I will give a derivational interpretation in terms of ‘cycles’ in section (2.3).

When elements are combined such that associativity does not hold, they take scope relative to each other. Depending on the bracketing, in the following structures ‘and’ has either wider scope than ‘or’ or more narrow scope.

- (39) a. A || or B | and C *or > and*
 b. A | or B || and C *or < and*

Within each level of embedding, the conjuncts are separated from each other by boundaries of *equal rank*. If any conjunct is itself complex and contains further nested bracketings, the boundaries within that conjunct have to be of *lower ranks*, and the operators within this domain take narrow scope relative to the operators at higher levels of embedding. The syntactic grouping imposes a ranking of boundaries. The proposed generalization about the grouping can be summarized as follows:

(11) SCOPALLY DETERMINED BOUNDARY RANK (**SBR**):

If Boundary Rank at a given level of embedding is n , the rank of the boundaries between constituents of the next higher level is $n+1$.

A level of embedding can be then identified both based on its prosodic and interpretative properties. It forms a prosodic domain and is set off with stronger boundaries from material outside of this domain; semantically, it forms a domain in which the associative law holds.

If each conjunction and disjunction is binary, as I have been tacitly assuming, then we need to refer to associativity in order to explain why sometimes prosody in coordinate structures is flat, and sometimes it is more articulated. There is an alternative to the binary view of connectors, however. What if ‘and’ and ‘or’ are functors that are able to take infinitely many arguments? In this case, structures with ‘flat’ prosody can be analyzed as projections of a single ‘and’ or ‘or’. We could then capture the differences in coordinate structure by stipulating that each single coordinate head defines a ‘level of embedding’ but can connect more than 2 elements in a prosodically ‘flat’ way—without reference to associativity. I will give some arguments against this view in the appendix in (2.5.1).

More generally speaking, the alternative to the the approach in terms of associativity is to simply stipulate which domains are ‘cycles’ or ‘phases’—e.g. in Chomsky (2001), a list of phasal heads is given that are proposed to be phasal.¹⁶ It is certainly

¹⁶According to Chomsky (2001), Nissenbaum (2000), it should be the *complement* of a phasal head that forms a cycle, so we would have to stipulate a head that takes each coordinate phrase as its complement, or show that every element that takes coordinate phrases as a complement happens to be phasal. It could of course be that sometimes coordinate phrases are *not* a separate cycle.

possible that it is necessary to stipulate a list of phases, and that there is no deeper generalization behind the list of cyclic domains. Using the associative law in the definition of a ‘cycle’ or ‘phase’ strikes me however as a more interesting possibility and the evidence from coordinate structures discussed here and from other domains discussed in chapter 3 suggests that it has some promise.

2.3 The Syntactic Cycle

The second step in the argument is to establish how the syntax of coordinate structures works. An important question is how many different bracketings there are in a coordinate structure with n conjuncts, and what representations of syntax are apt to encode them in an insightful way. A closer look at the syntax paves the way to link the observed prosody in a straightforward way to the syntactic derivation, exploiting the concept of the syntactic ‘cycle’.

2.3.1 The Combinatorics of Coordination

A theory of syntactic representation should give us just the right number of bracketings for phonological and semantic interpretation. How many bracketings are there in coordinate structures, and what theories of syntactic representations have the right combinatorial power?

We can use the methods to induce bracketing described in section 2.2 to establish how many bracketings are possible. Based on the findings so far, it seems that we have to distinguish at least three bracketings for the case of $n = 3$ conjuncts:¹⁷

- (40) a. Lysander | and Helena | and Demetrius.
 b. Lysander | and Helena || or Demetrius.
 c. Lysander || and Helena | or Demetrius.

¹⁷This is not to say that there are only three ‘readings’ of these coordinate structures. But all the strategies of forcing further bracketing discussed seemed to converge on one of these three prosodic options.

There is sufficient confusion in the literature on coordination to warrant a closer look at the combinatorics involved. Gleitman (1965, 276) gives 3 different bracketings of structures involving $n = 4$ conjuncts all connected by ‘and’; Langendoen (1998) claims that there are 7. How many bracketings are there for n -many conjuncts in general?¹⁸

If all coordinate structures were binary, and if there are only binary branching trees (as argued i.a. in Kayne (1983)), then it would be unexpected that there are *three* different syntactic structures with 3 conjuncts. There are only two binary branching trees that one can construct for three elements under these assumptions:

¹⁸The bracketing of the functors themselves is irrelevant and not used for syntactic disambiguation. There might be a contrast between cliticizing to the right or to the left, but this, as far as I can tell, does not correspond to a syntactic/semantic distinction:

- (1) a. Lysander and Hermia.
- b. Lysander’n Hermia.

In the following I will group the connectors with the following conjunct. There are several arguments in favor of this grouping. Ross observed the following contrast (Ross, 1967, 90–91):

- (2) a. John left. And he didn’t even say good-bye.
- b. * John left and. He didn’t even say good-bye.

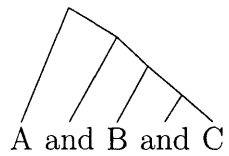
When a conjunct is extraposed, it is the second conjunct that extraposes, and the connector has to extrapose as well Munn (15 1993, , attributed to Gert Webelhuth):

- (3) a. John bought a book yesterday, and a newspaper.
- b. * John bought a newspaper yesterday a book and.
- c. * John bought a newspaper and yesterday, a book.

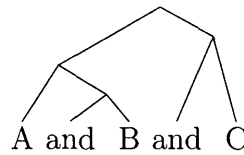
Since the connectors always group with the conjunct following them, we can ignore them when we count the syntactic bracketings of conjuncts in coordination structures.

(41) Two Binary Branching Trees, Assuming Binary Functors

a. Right-Branching



b. Left-Branching



The combinatorial question of how many binary branching trees there are over n linearly ordered elements is known as ‘Catalan’s Problem’.¹⁹ The number of binary branching trees over n linearly ordered items is the Catalan number of $(n-1)$. If coordination is always binary and attaches to the right, then the coordinators can be ignored in assessing the number of possible coordination with n conjuncts. The Catalan number for $n = 3$ is 2. The recurrence relation for the Catalan number is given below:

(42) Catalan number

Catalan Number of $(n-1)$ is the number of binary branching trees over n linearly ordered items.

$$C_{n+1} = \frac{2(2n+1)}{n+2} C_n$$

Catalan Sequence: for $n = 1, 2, 3, 4, 5, \dots$, C_n is 1 1 2 5 14 ...

While the Catalan number of $n = 3$ equals 2, the number of bracketings in coordinate structures is (at least) 3, as was shown above.²⁰

¹⁹Named after the Belgian mathematician Eugène Charles Catalan (1814-1894). For references to proofs to the solution to Catalan’s problem see Sloane and Plouffe (1995) or Sloane’s archive of integer sequences at <http://www.research.att.com/~njas/sequences/Seis.html>. The Catalan sequence is Sloane’s A000108.

²⁰Geoff Pullum claims in a language log post that there are 3 bracketings for 3 nominals: ”This corresponds to the fact there are three bracketings for lifestyle consultation center: [lifestyle consultation center], [[lifestyle consultation] center], and [lifestyle [consultation center]]”. <http://itre.cis.upenn.edu/~myl/language-log/archives/000160.html>. Thanks to Chris Potts for pointing this out to me.

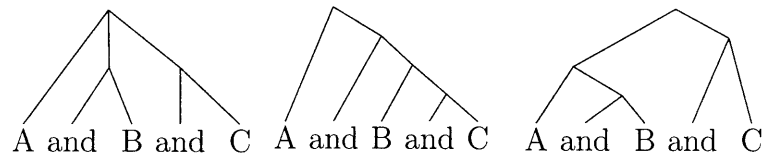
This is true at least if all connectors are identical, since if we alternate and/or the flat structure is ruled out. Only the two articulated prosodies are needed. Only for this case does the Catalan sequence gives us the correct result. Church and Patil (1982) show that this number of bracketings is the one generated by a phrase structure rule that assumes each step of coordination to be binary:

$$(43) \quad A \rightarrow A \text{ Co } A$$

One way to get to the right number of structures is to abandon the assumption of binary branchingness:

(44) Trees with Unbounded Branching

a. Flat b. Right-Branching Left-Branching



This is combinatorially equivalent to positing phrase structure rules that allow any number of arguments (Chomsky and Schützenberger (1963)):²¹

$$(45) \quad A \rightarrow A \text{ Co } A (CoA)^*,$$

where $(Co A)^*$ is zero or more occurrences of $Co A$

The number of bracketings for n conjuncts is then the number of trees over linear strings allowing arbitrary branching. This number is called the Super-Catalan number of n (see Stanley (1997) for discussion and references to proofs).²²

These numbers are also called ‘little Schröder numbers’ and ‘Plutarch numbers’.²³

A definition in terms of a recurrence function is given below:

²¹See also Gazdar et al. (1985).

²²The Super-Catalan Sequence is Sloane’s A001003.

²³Stanley (1997) gives an incisive introduction into the combinatorics of this sequence and the history of its discovery, from which the following information is distilled:

Plutarch (ca. 50 A. D.–120 A. D.) was a Greek biographer. In his *Moralia*, Plutarch states the following: “Chrysippus says that the number of compound propositions that can be made from only ten simple propositions exceeds a million. (Hipparchus, to be sure, refuted this by showing that on the affirmative side there are 103049 compound statements, and on the negative side 310952.)”

(46) Super-Catalan Number

Super Catalan Number of n is the number of unbounded branching trees over n linearly ordered items.

$$S_n = \frac{3(2n-3)S_{n-1}(n-3)S_{n-2}}{n}$$

Super-Catalan Sequence: For $n = 1, 2, 3, 4, 5, \dots$, S_n is 1 1 3 11 45...

The possible bracketings of coordinations with $n = 3$ and $n = 4$ are summarized schematically below:

- (47) a. $n = 3$
xxx, x(xx), (xx)x
- b. $n = 4$
xxxx, xx(xx), x(xx)x, (xx)xx, x(xxx), (xxx)x, (xx)(xx), x(x(xx)),
x((xx)x), ((xx)x)x, (x(xx))x

In the previous section several ways of inducing prosodic bracketings were presented. We can use the bracketings in (47) and try to replicate them with prosodic bracketing in coordinate structure. We can induce all of the bracketings in (47) by keeping the functors within a level of embedding constant.

Changing the functor from ‘and’ to ‘or’ or vice versa will induce a hierarchical organization, as discussed above. It thus forces a prosodically articulated structure

103049 is the Super-Catalan of 10, and is the number of bracketings for $n=10$ conjuncts. If we only combine propositions, as Plutarch suggests, then the associative law would render bracketing irrelevant in cases where we iterate the same connector. If we allow ourselves the use of both and/or, then the number of complex statements is of course $2 \cdot 103049$, since for each bracketing with ‘and’ at the top level (of which there are Super-Catalan-many) there is one with ‘or’ at the top level, so Hipparchus may have been off by a factor of 2, or Plutarch quoted him wrongly.

See Stanley (1997) for a discussion of the actually fairly recent discovery that Plutarch was not just throwing out some random digits, in 1994, and Habsieger et al. (1998) for an interpretation of Plutarch’s second number.

Ernst Schröder (1841–1902) was the first mathematician to specify a generating function.

precisely at the points where in orthography we write parentheses. In this way we can create structures corresponding to each of the ones listed in (47). This means that there are at least Super-Catalan-many bracketings in coordinate structures. To illustrate the point, consider first the case of $n = 3$.

The three prosodic bracketings of three elements in are illustrated in (48). (48a) is a disjunction with three alternatives; (48b) is a disjunction with two alternatives, the first of which is internally complex; (48c) is a disjunction with two alternatives, the second of which is internally complex. We alternate the functor exactly at the point that corresponds to parentheses in (47a):

- (48) a. Lysander | or Hermia | or Helena.
 b. Lysander | and Hermia || or Helena.
 c. Lysander || or Hermia | and Helena.

Let's now turn to coordination structures with $n = 4$ proper names. The Super-Catalan number for $n = 4$ is 11. The simplest structure is a simple coordination or disjunction of four elements. The example given here is a disjunction of four alternatives:

- (49) | Lysander | or Demetrius | or Hermia | or Helena. |

There are more complex structures to be considered, however. Consider the following three structures, all of which semantically involve a disjunction with three alternatives, one of which is internally complex:

- (50) a. Lysander || or Demetrius or || Hermia | and Helena.
 b. Lysander || or Demetrius | and Hermia || or Helena.
 c. Lysander | and Demetrius || or Hermia || or Helena.

The following are structures with two alternatives, one or two of which are internally complex:

- (51) a. Lysander || or Demetrius | and Hermia | and Helena.

- b. Lysander | and Demetrius | and Hermia || or Helena.
- c. Lysander | and Demetrius || or Hermia | and Helena.

There are four more structures which involve two alternatives. Just as in (51a,b), one of the two is internally complex. The difference to the structures in (51a,b) is that the complex structure involves another level of embedding:

- (52)
- a. Lysander ||| or Demetrius || and Hermia | or Helena.
 - b. Lysander ||| or Demetrius | or Hermia || and Helena.
 - c. Lysander || and Demetrius | or Hermia ||| or Helena.
 - d. Lysander | or Demetrius || and Hermia ||| or Helena.

8

Structures with two levels of embedding are reported from corpora in Langendoen (1998, 240).²⁴

- (53) Combine grapefruit with bananas, strawberries and bananas, bananas and melon balls, raspberries or strawberries and melon balls, seedless white grapes and melon balls, or pineapple cubes and orange slices. (From the APHB corpus).

Langendoen (1998, 243) claims that two levels of embedding are impossible if the functors are all identical, and claims that in this case that there are only seven structures. This, however, seems to me to be incorrect.

Consider a situation in a diner, in which two guests have ordered breakfast. One guest ordered the special, which consists of bacon and eggs and a side of salad; the other guest ordered french toast. After a change of shift, a new waiter asks what was ordered:

²⁴Langendoen (1998, 241) states as a fact that coordinate structures with n levels of embedding, with $n > 2$, are ungrammatical—this would mean that the depth of embedding of the structures in (52) is the maximal depth that natural language can generate—but he does not present any evidence in favor of this claim, apart from the intuition that a fully articulated right-branching structure involving only ‘or’ sounds odd out of context, which is hardly surprising.

- (54) What should I take to tables 1 and 2 respectively?
 ((bacon and eggs) and salad)) and French toast

Again, this may not be the most practical way of communicating, but it seems to me that it is quite possible to use these answers, and that there is a grouping involved—least at the semantic/syntactic level. Whether or not the levels of embedding can all be prosodically differentiated is an empirical question. The question of how prosody differentiates structures with two levels of embedding is discussed in chapter 5.

The boundary ranks for the eleven structures containing four elements are summarized below:

- (55) Scopally Determined Boundary Ranks

							A	B	C	D
1	A	or	B	or	C	or	D	1	1	1
2	A	or	(B	and	C)	or	D	2	1	2
3	A	or	B	or	(C	and	D)	2	2	1
4	(A	and	B)	or	C	or	D	1	2	2
5	(A	and	B)	or	(C	or	D)	1	2	1
6	(A	and	B	and	C)	or	D	1	1	2
7	A	or	(B	and	C	and	D)	2	1	1
8	((A	or	B)	and	C)	or	D)	1	2	3
9	(A	and	(B	or	C))	or	D	2	1	3
10	A	or	((B	or	C)	and	D)	3	1	2
11	A	or	(B	and	(C	or	D))	3	2	1

Natural language coordinate structures thus have at least Super-Catalan-many bracketings. I will assume from now on that there are *exactly* Super-Catalan-many bracketings.

The combinatorics presented in this section imposes a minimum on the number of structures that the grammar should provide for coordinate structures. A notation only allowing for binary branching trees, it seems, is not sufficient to distinguish enough coordinate structures.

A closer look at the syntax of coordinate structures offers an isomorphic alternative to flat trees that involves binary branching trees after all.

2.3.2 Syntactic Asymmetries in Coordination

In the exposition of the different possible coordinate structures over n elements in the preceding section we found that using unbounded branching introduces the right number of different tree structures, namely a number corresponding to the Super-Catalan number of n . One of the bracketings of n elements involves a ‘flat’ structure, i.e. a structure that semantically involves iteratively combining n elements with either ‘and’ or ‘or’. This structure is prosodically ‘flat’ in that the conjuncts are separated by boundaries of equal rank and are thus prosodically on a par.

‘Flat’ unstructured representations have sometimes been taken as the null hypothesis for coordinate structures. Miller and Chomsky (1963, 196) take this to be the obviously correct approach: “Clearly, in the case of true coordination, by the very meaning of this term, no internal structure should be assigned at all within the sequence of coordinate items.”

Unbounded trees were assumed in many subsequent generative approaches, e.g. Gleitman (1965) combines an arbitrary number of conjuncts under and n -ary branching single node and speaks of a ‘main’ and ‘subordinate’ conjunction in cases of nesting of coordinate structures that do not have a flat prosody.

This section presents evidence, however, that while the structures in question may be ‘flat’ in a prosodic sense, syntactically there is a clear asymmetry between the conjuncts. The source of evidence for the asymmetry is variable binding. Consider semantic binding first (Munn, 1993, 16):

- (56) a. Everyone_{*i*} and his_{*i*} goat was at the party.
b. * His_{*i*} goat and everyone_{*i*} was at the party.

Semantic binding is possible from the first conjunct into the second but not vice-versa.²⁵ There is a clear asymmetry in coordination structures. But what is the

²⁵Progovac (2003) argues that variable binding is licensed not by surface c -command but at LF after raising out the quantifier. The following example illustrates that this view is too simplistic:

- (1) a. Every student_{*i*}, his_{*i*} adviser, and the dean are invited to the reception.
b. * His_{*i*} adviser, every student_{*i*}, and the dean are invited to the reception.

exact generalization about when semantic binding is possible?

Based on the examples so far, it could be that in coordination structures, linear precedence is sufficient for variable binding and for inducing condition-C effects. But more has to be said:²⁶

- (57) a. Every student_i and (his_i parents or his_i adviser) have to be at the graduation.
- b. * (Every student_i and his_i parents) or his_i adviser have to be at the graduation.

This sentence does not have the reading indicated by the brackets in (b), in which it would be sufficient if all the advisers would show up, but no student. It seems that it is impossible to bind a variable from a more embedded conjunct into a conjunct of the next higher coordination or disjunction.

The left-right asymmetry persists even when the second connector takes scope over the first connector:

- (58) * (His_i adviser and his_i parents) or every student_i have to be at the graduation.

There is a simple generalization that captures the data—at least if we assume the binary right-branching representation of ‘flat’ coordinations:

The movement necessary out of the conjunction in (1) example would violate Ruys’ generalization about movement out of conjunctions. The following examples illustrate further complications in the data pattern:

- (2) a. ? The dean, every student_i, and his_i adviser are invited to the reception
- b. * Every student_i, the dean, and his_i adviser are invited to the reception.

I am not sure how to account for the contrasts in judgements. Maybe Schein (2001)’s proposal that variable binding in coordinate structures involves ‘telescoping’ can account for them.

²⁶It is not clear whether (a) involves genuine variable binding or involve e-type variables. Negative quantifiers could rule out an e-type reading, and it seems that they lead to ungrammaticality, at least with an articulated structure. The contrast between (57a,b) has to be accounted for in any case, and it suggests a distinction in terms of c-command.

(59) C-command

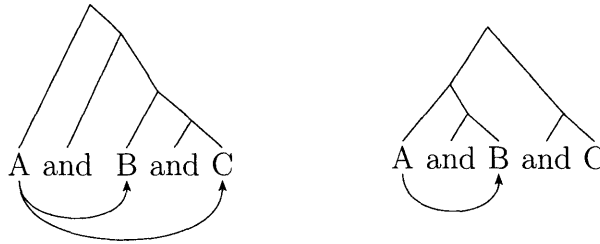
A node A c-commands a node B, if and only if A's sister either

- a. is B
- b. or contains B.

This is the definition of c-command used in Adger (2003a, 117).²⁷ It correctly distinguishes left- and right-branching structures. Consider the following tree representations. All c-command relations between the conjuncts A, B, and C are represented by arrows:

(60) Left- *vs.* Right-Branching and C-Command

- a. Right-Branching
- b. Left-Branching



A c-commands C in the right-branching but not in the left-branching structure. In the binary-branching approach, even the coordination structures with a ‘flat’ prosody would have a right-branching tree. The structure in (57b) is ungrammatical because ‘every student’ does not c-command the variable. It is further embedded in a left-branch and the variable is not within the sister of the quantifier phrase.

Of course, the entire node (A and B) in the left-branching structure c-commands C, and consequently it can bind a variable within C:

(61) (Every student or visitor)_i and his_i parents are invited.

The important point is that A and B themselves do not c-command constituent C. Variable binding thus constitutes evidence that the structures that have a ‘flat’

²⁷It is close to the standard definition of c-command, at least assuming bare phrase structure Chomsky (1994); Reinhart (1976, 23) proposed a more elaborate definition, in part due to the possibility of intermediate projections.

prosody actually have the same c-command properties as prosodically articulated right-branching structures.

The observed pattern is compatible with an analysis that assumes binary branching, takes ‘flat’ coordinate structure to involve right-branching structures and adopts the standard notion of c-command. But unbounded branching gives us the correct combinatorial result for coordination structures with n elements, and thus seem necessary based on the combinatorial results. How can we square these conflicting results?

2.3.3 Binary Branching Revisited

In the following, I will present a notation using labeled binary branching trees that is isomorphic to the representation of flat trees. Since the two notations are isomorphic in their combinatorial power, we are free to choose based on some other consideration. In the interest of saving a unified definition of c-command without reference to linear precedence, I will choose the binary branching notation—but this is a choice of taste and not of empirical argument.

There is a translation of flat trees into binary branching trees. It is obvious that the set of binary branching trees is a subset of the set of arbitrarily branching trees. We can also give a translation of an arbitrarily branching tree into a binary tree (cf. Rogers (2003)). The idea is to use the linear precedence relation in the flat branching nodes (e.g. $A \prec B \prec C \prec D$) and convert it into a nested right-branching structure (i.a. $[A [B [CD]]]$ in this case). Each flat tree is assigned a unique binary branching tree in this way.

The translation of flat trees into binary trees also does not have to affect the way the composition of meaning proceeds.²⁸ In fact, the compositional interpretation of the translated structures does not differ at all from the way the meaning of the flat structures are composed if one follows Buring (2005, 40) in his definition of functional application for flat structures. Instead of using binary syntactic sisterhood, Buring

²⁸Semantically, a function that takes n elements can be ‘Schönfinkelized’ or ‘curried’ into separate unitary functions (cf. Heim and Kratzer, 1998), so there is no reason to suspect that a translation of a flat tree into a binary branching tree would require a different theory of semantic composition.

uses the functor-argument asymmetry of adjacent constituents. For example, he composes flat VPs with two arguments by combining the predicate successively with its arguments one by one following linear order. The denotation of adjacent constituents are combined in a pairwise fashion. Function application and variable binding of flat structures then simply mimic the compositionality of a binary-branching structure. The translation to right-branching structures guarantees that it is always two uniquely defined sisters that compose and this voids the need to refer to adjacency.

The translation from flat trees to binary trees described here is not reversible: by converting flat branching nodes into right-branching we ‘neutralize’ the distinction between ‘originally right-branching’ and ‘originally flat’ nodes. What we need in order to show that we can represent all relevant structures with binary branching trees is an isomorphism that allows us to go from the arbitrarily branching trees to binary branching trees and back without loss of information. For this we have to annotate the binary-branching trees—eventually, we can abandon this annotation and replace it by a cyclic theory of syntax.

The piece of information that is lost is that all sisters in a flat branching tree are dominated by the same node. This missing information can be added to the binary trees by introducing labels to the tree notation. Borrowing from the notation tools of GB theory, we can label the nodes in the original flat tree ‘XP-nodes’. The nodes newly introduced in the conversion from flat to right-branching remain unlabeled (we can refer to them as $\bar{X}$ nodes).

A property of the conversion function is that ‘unlabeled nodes’ or ‘ $\bar{X}$ ’ nodes are only assigned to right-branching nodes. We can call all items combined under a single XP-node (with intermediate $\bar{X}$ -nodes) $\bar{X}$ -shells or ‘extended projections’. Larson (1988) indeed proposes extended projections to be right-branching, at least for VP-shells in English. Haider (1993, 2000) conjectures that this is a systematic property of all extended projections and states it as follows, essentially requiring a right-branching structure for extended projections (or shell-structures):²⁹

²⁹At least two syntactic approaches have in some way or other built into the system a preference of right-branching over left-branching structures: Kayne (1994) and Haider (1993). A detailed

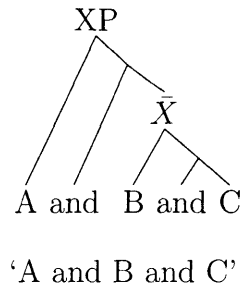
(62) Branchingness Conjecture (Haider, 2000)

For any two nodes *directly* attached to the *same* (extended) projection line, the node that precedes c-commands the node that follows.

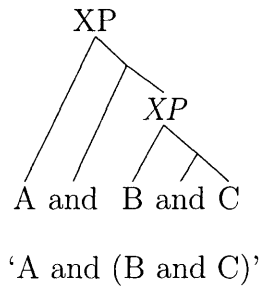
Imagine that we would also allow left-branching $\bar{X}$ shells. Then for $n = 3$, if we allow an $\bar{X}$ label for any branching node (except the root) we get too many different structures, e.g. for $n = 3$ we get 4 different structures instead of 3:

(63) Right-Branching

a. Flat prosody



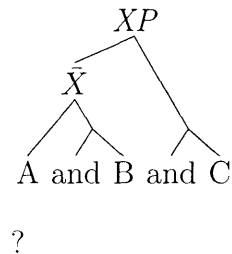
b. Hierarchical Prosody



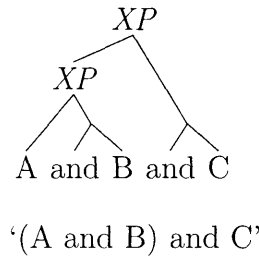
(64) Left-Branching

a. Flat Prosody

(Non-existent?)



b. ‘Hierarchical Prosody



If we stipulate right-branching for $\bar{X}$ shells, labeled trees provide a combinatorial system consisting only of binary branching trees that map one-to-one to arbitrarily branching trees and thus a system that encodes Super-Catalan-many bracketings. The proposed mapping function has several useful properties. Once we convert the consideration of coordinate structures within those two systems would go beyond the scope of this chapter.

flat structures according to the mapping function, there are exactly 2 right-branching structures for $n=3$ conjuncts, one prosodically ‘flat’ and one prosodically articulated:

- (65) a. Lysander | or Demetrius | or Hermia.
- b. Lysander || or Demetrius | and Hermia.

On the other hand, there is only a single left-branching structure, and this structure is prosodically fully articulated:

- (66) Lysander | or Demetrius || or Hermia.

The right-branchingness of $\bar{X}$ -shells in the binary notation leads to the generalization that right-branching structures can be prosodically flat, but left-branching structures are necessarily prosodically articulated.³⁰

- (67) Left-Branching Structures are Prosodically Articulated.

There are then two isomorphic ways to achieve the correct combinatorial power:

- (68) Two ways get Super-Catalan-many structures
 - a. Unbounded Branching trees.
 - b. XP vs. $\bar{X}$ distinction in label of nodes + Right-Branching of Shells

They are notational variants of each other, at least with respect to the number of phrase markers/derivations they provide for a tree with n terminal nodes. The decision between them should be guided by the heuristic of which of those options allows for the simplest statement of relevant generalizations. The important lesson to learn from the combinatorics of coordination is that whichever of the two options in (68) we employ, we should pick *exactly one* of them, since otherwise the combinatorial possibilities explode.

Labeling nodes as XP -nodes or $\bar{X}$ -nodes allowed us to use binary branching trees, and to maintain the standard notion of c-command—at the price of having to stipulate that shell-structures are right branching. The next section suggests a more insightful derivational view.

³⁰Possessors in English seem to be a counterexample. See chapter 3 for discussion and evidence that they are underlyingly right-branching.

2.3.4 The Syntactic Cycle

This section presents a derivational view of syntax that will allow us to get rid of the labels (XP *vs.* $\bar{X}$) and has the desirable property that it allows us to exploit something that any generative theory assumes in order to define what the notion of a level of embedding is: the recursiveness of language.

The derivational theory distinguishes nodes that undergo a spell-out, and thus form a separate work-space in the cyclic building up of structure, from nodes that do not, and are part of a bigger structure constructed in a single work-space. Instead of using labels, the nodes we labeled XP -nodes earlier to constitute cycles, while those that are $\bar{X}$ -nodes do not.

This allows for a simpler statement of the right-branching conjecture:

(13) Right-Branching Conjecture

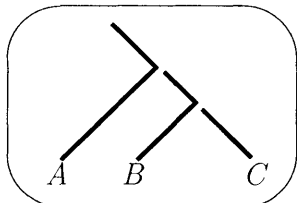
Each cycle consists of a right-branching structure.

This is similar to the principle ‘Branch Right’ proposed in Phillips (1996), which favors right-branching structures unless it is not compatible with the semantic interpretation of an expression. The main difference is a technical one, namely that I am assuming a cyclic theory of syntax and stipulate right-branchingness as an necessary property of every syntactic cycle.

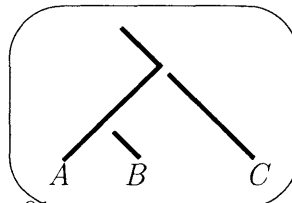
The idea is that a right-branching structure is the *only* structure the grammar can assemble given a numeration of elements. I will call the domain in which structures are assembled a cycle (workspace/phase/level of embedding). Within any workspace, only (69a) can be generated, (69b) is impossible:

(69) Structure within a Cycle

a. Possible:



b. Impossible:



In the absence of a motivation for either a left- or right-branching bracketing, grammar imposes a right-branching analysis. This is reflected in the preference for right-branching structures, which were developed into the asymmetric syntactic theories in Haider (1993), Kayne (1994), and is related to the ‘cascade’ structure discussed in Pesetsky (1995).

It is still possible to assemble left-branching structures—just not within a single cycle. The additional assumption that I make use of is that of recursion:

(70) Recursion

Each output of a cycle can enter a new cycle as an atomic expression.

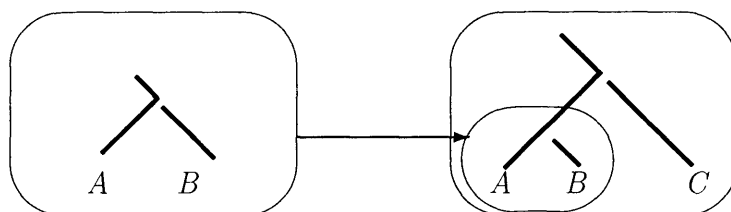
This assumption is implicit in all syntactic theories I know of. What is not shared by all theories is that the derivational steps in the recursion matter in capturing grammatical phenomena. I will take this derivational view of syntax, as in (Uriagareka, 1999, Johnson, 2002).³¹

Left-branching structures require the combination of more than one work-space, while right-branching structures can be created in one go. Consider the following derivation:

(71) Creating a Left-Branching Structure

a. First Cycle

b. Second Cycle:



³¹Johnson (2002) uses a notion ‘Re-numeration’, which is close to the way I will use the cycle, in order to explain adjunct island effects and also focus projection effects.

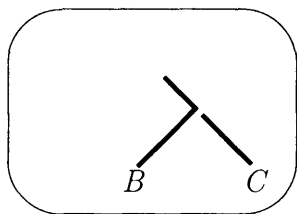
This is the *only* way of deriving a left-branching structure. A right-branching structure might be derived either in one workspace, or in two steps. There are thus 3 ways of assembling structures with 3 elements. In fact, for n elements, there are exactly Super-Catalan of $(n-1)$ many ways of building structure. We can simply identify right-branching structures built in a single cycle with the flat nodes of the arbitrarily branching trees or with the shell-structures of the labeled tree representation.

The cycle has also been claimed to have an interpretative property, namely that the elements combined in it are combined in such a way that the associative law holds. In other words, we do not need labels to identify the cyclic nodes, it suffices to look at interpretative properties.

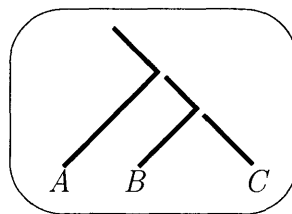
This means that the bracketing of the material within a cycle is immaterial. This property enables us to build up the structure in a cyclic way either top-down or bottom up. Derivations are conventionally assumed to work bottom-up:

(72) Bottom Up

a. First Merge



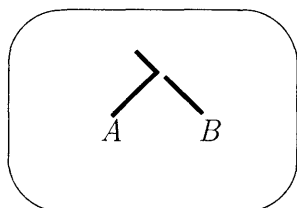
b. Second Merge



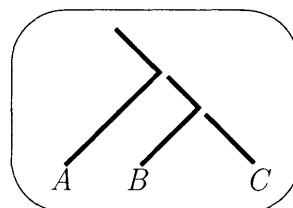
But this not the only possibility. Phillips (1996, 2003) proposed to build up structure left-to-right, in such a way that each step of Merge destroys phrase structure built up earlier:

(73) Top-down

a. First Merge



b. Second Merge



There is some evidence that a left-to-right, top-down derivation is possible in prosodically ‘flat’ coordinate structures. Frazier et al. (2000, 360) report results from an experiment in which subjects were asked to read coordinate structures, varying the number of conjuncts and whether or not all or only the last coordinator was pronounced.

- (74) a. John and Fred.
b. John, (and) Bill and Fred.
c. John, (and) Bill, (and) Fred, and Harry.

There was no effect on reading time due to the number of conjuncts (or the presence/absence of connectors). This suggests that material pertaining to a single cycle can be assembled from left to right by subsequently adding new constituents, rather than assembling the structure bottom up. Otherwise, the processing cost of the ever increasing number of upcoming constituents should affect the realization of the preceding constituents. This would be a useful property of a unit for language processing in production and comprehension.

For the structure built within a single cycle, it is immaterial which of the options (72) or (73) we choose. If cycles are associative domains, then they are robust with respect to directionality of composition.

In the following I will assume then with Phillips (1996, 2003) that the order of the derivation is left-to-right, and the order in which elements are spelled-out into a new cycle is directly reflected in their linear order.

The last and final step of the argument is to show that the cyclic approach to syntax promoted here gives an elegant account of how to derive the prosodic structure.

2.4 Prosody and the Cycle

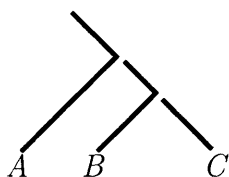
The last step in the argument is to show how the cyclic syntactic derivation can be put to use to derive a metrical grid that encodes the correct prosodic structure.

2.4.1 Prosodic Representation

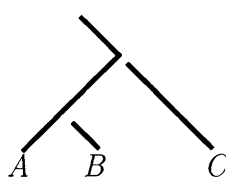
Pierrehumbert (1999) defines prosody as the "grouping and relative prominence of the elements making up the speech signal". An adequate representation of prosodic structure should reflect intuitions about grouping, should transparently encode prosodically relevant information such as boundary strength and accent placement, and should encode prominence relations. Boundary strength is not transparently encoded in the syntactic tree structures usually employed in generative syntax:

(75) Syntactic Bracketing and Boundary Strength

a. A or (B and C)



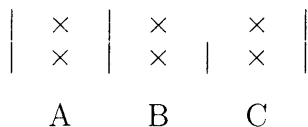
b. (A or B) and C



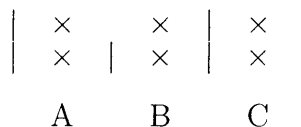
I will therefore use the notation of the *metrical grid*, a notion first proposed by Liberman (1975). The metrical grid reflects the timing structure of an utterance, and was originally proposed in analogy to musical notation. The particular notation I will use here is a version of the bracketed metrical grid (cf. Halle and Vergnaud, 1987). I illustrate the notation used here:

(76) Metrical Grid

a. A or (B and C)



b. (A or B) and C



The notation is to be read as follows: The unoriented parentheses ('pipes') demarcate feet at each level in the grid. Foot boundaries at higher boundaries necessarily correlate with foot boundary at lower grid lines. And a foot can contain multiple grid marks, i.e. feet are unheaded.

The notation diverges from the one employed in Idsardi (1992) and Halle and Idsardi (1995) in using foot boundaries that are not directional (left/right

parentheses). Grids with undirected brackets have been proposed in Hall (2000) and Reiss (2003).³² A further difference is that I am using matching parentheses, while Idsardi (1992) proposed unmatched parentheses—the difference is again not crucial for the data-set I am discussing here.

The representation encodes boundary strength in a straightforward way: feet at higher grid lines have stronger boundaries. The boundary rank can now be read off as the height of the column of pipes.

(77) BOUNDARY RANK

The rank of a prosodic boundary is the number of the highest grid line containing a pipe at that boundary.

How do boundaries induce the lengthening effects that at the heart of speakers' intuition about boundary strengths? One possibility is that the boundaries represented by the 'pipes' actually have a direct gestural correlate, which affects the timing of the realization of the segmental score. Such a boundary gesture was proposed in Byrd and Saltzman (1998) and Byrd and Saltzman (2003), the idea being that this prosodic gesture effects a slowing down immediately preceding the boundary. This effect recedes after the boundary, and hence there is a speed-up. I discuss durational evidence for boundaries in more detail in chapter 4 and chapter 5.

A second issue is the question of how does the representation encode the location of pitch accents in a structure? I will assume that the following generalization holds about the implementation of the metrical structure:

(78) ACCENT PLACEMENT

All and only top-line grid marks receive pitch accents. All other material is either unaccented or the accents are realized within a reduced pitch range.

In coordinate structures, in general each conjunct receives an accent, which is reflected in the notation here by projecting top line grid marks within each conjunct to the

³²Hall (2000), Reiss (2003) present evidence that they are sufficient to account for variation in stress systems. Since I am not concerned here whether or not the notation is powerful enough to capture variation in word stress systems, I will not discuss the issue in any detail.

top line of the entire structure.

The grid notation in (76) encodes information about grouping that is already present in the tree-structure in (75). In fact, it seems that it encodes the same information, albeit in a different way. This redundancy is not problematic as long as we conceive of the grid notation and the syntactic tree notation as two ways of looking at the same information. We can think of the metrical grid as *a way of drawing a syntactic tree and the history of how it was assembled*.

The version of the grid I employ here departs from earlier literature (Prince, 1983, Halle and Vergnaud, 1987, Idsardi, 1992) in a number of ways. One major difference is that it does not represent nuclear stress, i.e. it does not encode the syllable that is perceived by native speakers as the most prominent. Just as the originally proposed grid in Liberman (1975), Libermann and Prince (1977), it encodes the major ‘beats’, but does not single one of them out as being the main stress by projecting it to a higher line than any other beat in the string.

How then does the grid representation encode prominence relations? I assume that the notion of ‘nuclear stress’ can be derived from the present notation without further addition or projection to a higher line. Newman (1946) and Truckenbrodt (1995) observe that in English, main prominence is perceived on the last accented element within a constituent.

(79) NEWMAN’S GENERALIZATION (Newman, 1946, 176)

”When no expressive accents disturb a sequence of heavy stresses, the last heavy stress in an intonational unit takes the nuclear heavy stress.”

Coordination structures in English generally have main prominence within the last conjunct, but all conjuncts receive an accent. This is completely independent of the internal bracketing of the coordinate structure. In the following example, within the subject, the main prominence is perceived on ‘Helena’, no matter which of the bracketing of the coordinate structure is chosen:

- (80) a. Lysander and (Hermia or Helena) are in the forest.
b. (Lysander and Hermia) or Helena are in the forest.

In SPE, for every phrase marker the nuclear stress rule (NSR) identifies exactly one vowel as most prominent. The vowel that native speakers perceive as most prominent in a given structure counts as its nuclear stress. The representation of stress in SPE is that of a phonological feature that can have any positive integer number as its value. The nuclear stress of an utterance is the unique vowel that has the value 1. Within every sub-constituent, there is exactly one vowel that has the lowest number. We can call this the vowel carrying nuclear stress within that sub-constituent.

A different notion of ‘nuclear stress’ or ‘nuclear accent’ is used in the intonation literature. Newman (1946, 174) singles out a heavy stresses as ‘nuclear’ if it ‘acts as the nucleus of an intonational unit’. Within any intonational unit it is always the last heavy stress that counts as the nuclear stress. Newman (1946, 176) suggests that in coordinate structures (‘enumerations’) each conjunct receives a nuclear accent. The intuition that each conjunct is on a par will be captured here by the fact that the boundaries that separate them are identical. For the course of this section, I will stick to the SPE notion of nuclear stress, and I will return to Newman’s idea about nuclear stress in the next section.

The reason the notation in (76) already encodes where nuclear stress falls within each constituent is that it is always the *last accented syllable*. Under the assumption that it is all and only elements that project to the top grid line that get linked to a pitch accent, we can capture the generalization as follows:

(81) Nuclear Stress Generalization

Within each foot, nuclear stress is perceived on the last of those grid marks that project highest.

Since each conjunct projects to the top line and receives an accent, the examples so far are not a particularly interesting illustration of Newman’s generalization. Each conjunct in a coordinate structure projects, so nuclear stress is always final. In fact, the notation seems redundant, since we could just as well have used the following notation:

(82) Flat Notation

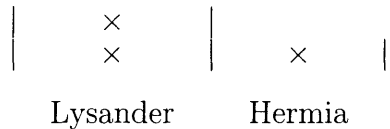
- | | |
|---|--|
| <p>a. Right-Branching</p> <p style="text-align: center;"> × × × </p> <p style="text-align: center;">A B C</p> | <p>b. Left-Branching</p> <p style="text-align: center;"> × × × </p> <p style="text-align: center;">A B C</p> |
|---|--|

But under certain circumstances some conjuncts do not receive an accent:

- (83) a. Demetrius and Hermia?
- b. No, Lysander and Hermia.

In a natural rendition of (83b) the material following the first conjunct is deaccented or at least heavily pitch reduced. It is *prosodically subordinated*. It is this type of prosodic subordination which is responsible for cases in which nuclear stress does not fall on the final element in a phrase. The conditions under which prosodic subordination applies and the generalization about nuclear stress will be discussed in detail in chapter 6 and chapter 7. We can present prosodic subordination in the grid as follows:

- (84) Subordination

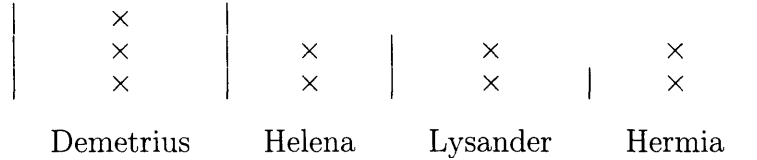


The ‘nuclear’ stress now falls on the syllable with main stress in ‘Lysander’, since it is the last unsubordinated element, i.e. the last element that projects to the top line. An important property of subordination is that the prosodic phrasing in the subordinated domain is maintained. Consider the following dialogue:

- (85) Who will get married?
- a. (Egeus and Helena) and (Lysander and Hermia)?
- b. No! (Demetrius and Helena) and (Lysander and Hermia).

The grid notation of the response looks as follows:

- (86) Phrasing in Post-nuclear domain



Nuclear stress of the entire answer is perceived on the main stress of ‘Demetrius’, since this syllable is the last (and only) one that projects a grid mark to the top line. For the same reason, nuclear stress within the first conjunct (‘Demetrius and Helena’) is perceived on that syllable.

The prosodic phrasing within the subordinated material is maintained, the break after ‘Helena’ is intuitively stronger than that after ‘Lysander’. That post-nuclear phrasing still reflects differences in boundary rank was experimentally tested in Ishihara (2003) and Jaeger and Norcliffe (2005).

Finally, the nuclear stress generalization given here goes beyond Newman’s generalization and applies to constituents that are entirely unaccented: The main prominence within the subordinated conjunct ‘Lysander and Hermia’ is perceived on ‘Hermia’, since this word contains the last syllable projecting to the highest line *within that constituent*.

The notion ‘prominence’ is thus not uniformly represented in the metrical representation: among a series of elements that project to the same grid-line, it is the last one that is perceived as most prominent, but elements that only project to lower lines are less prominent even when they are at the phrase end. The tacit assumption is that prominence is not a primitive of the theory but emerges in the task of asking prominence judgements. The benefit of the present representation is that it allows us to capture the observed prosodic grouping, as well as intuitions about prominence.

2.4.2 Prosody and the Cycle

In order to map syntactic structure to prosody, I propose a cyclic mapping to prosody. Just as in SPE, prosody is derived here using a ‘transformational cycle’. I assume that the syntactic cycle corresponds to the phonological cycle. This hypothesis was first proposed in Bresnan (1971), and has recently gained more currency in the context of ‘phase’-theory (Chomsky, 2001), i.a. Adger (2001), Marvin (2002), Arregi (2002),

Wagner (2002a), Adger (2003b), Dobashi (2003), Ishihara (2003), Kahnemuyipour (2003, 2004) and Selkirk and Kratzer (2005).

The idea that I want to propose here is very simple: Elements are combined by Merge throughout the cycle. When an associative domain is completed, the cycle is ‘spelled out’. Part of ‘spelling-out’ is to map the content of a cycle to a prosodic unit in the phonological representation. In particular, a new top line to the grid is added, and each top line grid mark within any of the elements of the cycle is projected up to this new grid line.

(16) PROSODIC MATCHING

a. Concatenate

Concatenate the prosodic representation of the elements in the domain aligning their top lines and filling the columns where necessary.

b. Project

Create a new top-line grid line n by projecting all grid marks on line $n-1$, and mapping them into a single foot on line n .

In essence, this principle simply maps the output of the cycle to a single foot in metrical structure. ‘Concatenate’ plays a similar role in the present theory as the ‘stress equalization principle’ in Halle and Vergnaud (1987). It assures that constituents that are combined in a cycle start out on an equal footing in their prosodic representation.

The elements of a cycle can be terminal nodes. I assume for simplicity that they come in as vocabulary items with some metrical structure associated with them in their phonological representation.³³; but they may also consist of the output of other cycles and be internally complex.

In order to apply this principle, no knowledge is needed about constituency within the cycle, about the nature of the objects in it (which is a functor, which is an argument), or about the information structure in the context. All of these factors

³³I will not discuss the cyclic foot structure below the word in this dissertation. We can think of the cycle relevant for within-word phonology as syntactic cycles as well, as proposed e.g. in (Marvin, 2002).

affect prosody, of course. But their effect is indirect, and is due to how these factors interact with syntactic recursion. This is why much of this thesis is dedicated to unravel properties of the syntactic derivation, although its ultimate goal is to account for the generation of the appropriate metrical structures.

Each conjunct in a coordination structure, I assume, has already undergone a cycle, and is thus spelled-out before in a separate cycle. I will not discuss the functors at this point but treat coordination structures as if they consisted just of the conjoined elements.

I will not discuss how linear order comes about. I will assume that along with assigning prosodic structure, at each cycle also the linear order within the cyclic domain is fixed, following Fox and Pesetsky (2005), Ko (2005), Sabbagh (to appear) and Takahashi (2005).

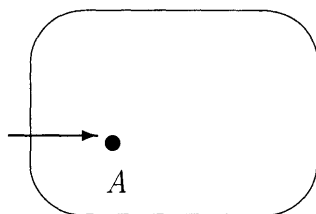
Merging a constituent in a cycle that already comes with a prosodic representation goes along with concatenating this element to the existing prosodic representation.³⁴

(87) Building a Right-Branching structure in one single cycle:

A | and B | and C

a. Start Point

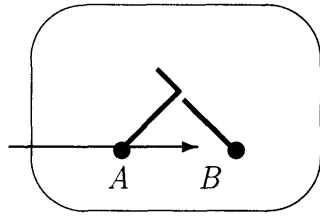
Prosodic Representation



| × |
A

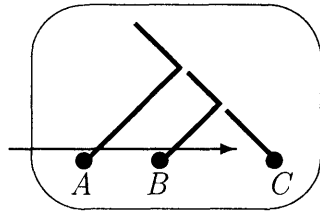
b. First Merge

³⁴I am presenting the derivation assuming with Phillips (1996, 2003) a top-down derivation. Given the discussion in section 2.3 above, it is not relevant whether a cycle is assembled bottom-up or top-down, and given the way the prosody is derived here it would not affect the outcome.



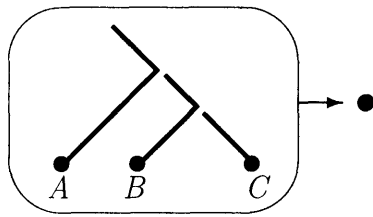
	×		×	
	A		B	

c. Second Merge



	×		×		×	
	A		B		C	

d. Spell-Out



	×		×		×	
	×		×		×	
	A		B		C	

Each of the terminal nodes in this tree could be terminal nodes³⁵ or themselves be cycles. In the coordinate structures discussed here they might always be separate cycles. Once spelled out, a cycle is opaque and the information within it is no longer accessible. *A filled circle stands in for a cycle that has already been spelled out.*

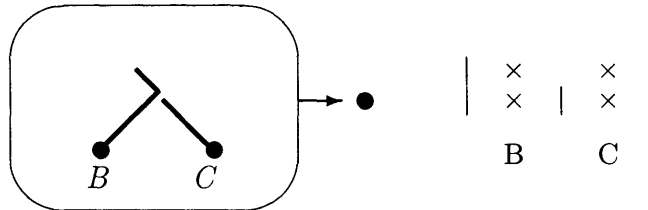
A right-branching structure with three terminal nodes can also be derived in two separate cycles. This becomes necessary when the associative law does not hold, as in the following expression:

(88) A and (B or C)

³⁵For simplicity we can assume that lexical vocabulary items come in with some metrical structure. I will not go into the syntactic cycle below the word in this thesis. One way of thinking about word internal cycles is to simply assume that they are also syntactic cycles (as in Marvin (2002)).

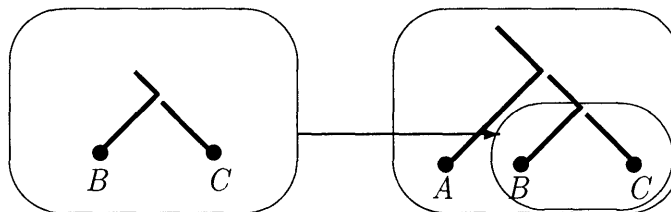
One cycle combines the sub-constituent (B and C).

(89) First Cycle, Spelled-Out



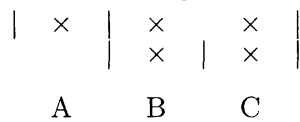
The output of the first cycle is inserted into a second cycle:

(90) Second Cycle



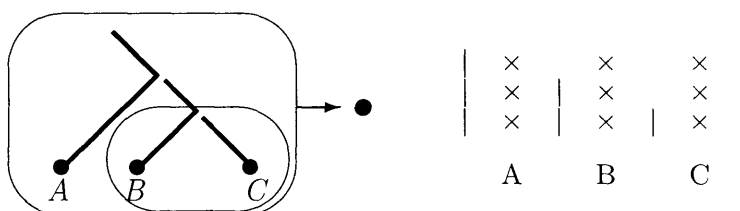
Before projecting the a foot on a new top grid line, the metrical elements corresponding to the elements in the cycle are concatenated. The grid representation of the concatenated grids looks as follows:

(91) Concatenating the Grids



When the second cycle is spelled out, a new grid line is projected. I will also 'fill up' the grid column that is unfilled, simply for notational reasons—the relations in the grid are relative, and constituents of different complexity can be matched to the same grid level.

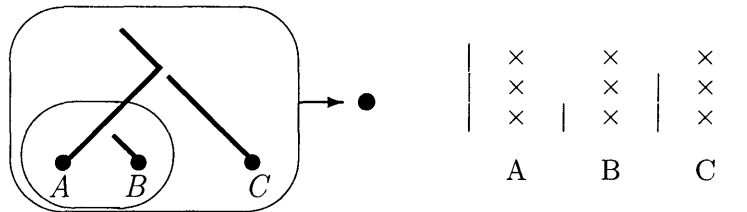
(92) Spell-Out of Second Cycle



Two different grids are derived, although the same binary tree is involved. The grid keeps track of the difference in derivation.

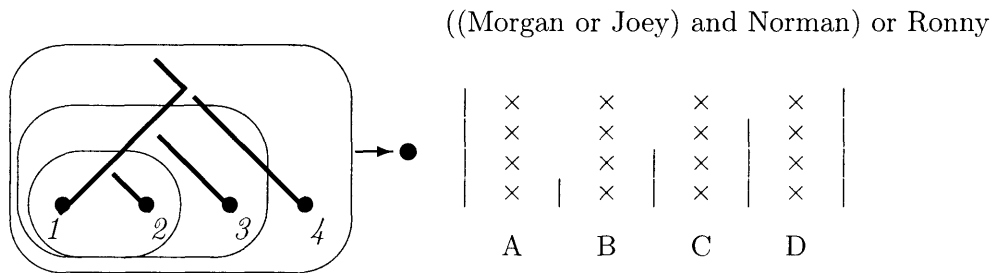
A left-branching structure necessarily is created via two cycles, as observed above. The grid that is derived looks as follows:

(93) Left-Branching Structure



Three different grid structures are derived, for the three different derivations. Similarly for the case of $n=4$, the metrical grid distinguishes 11 different derivations for the 5 different binary trees. Consider first a fully left-branching tree, which necessarily goes through 3 cycles:

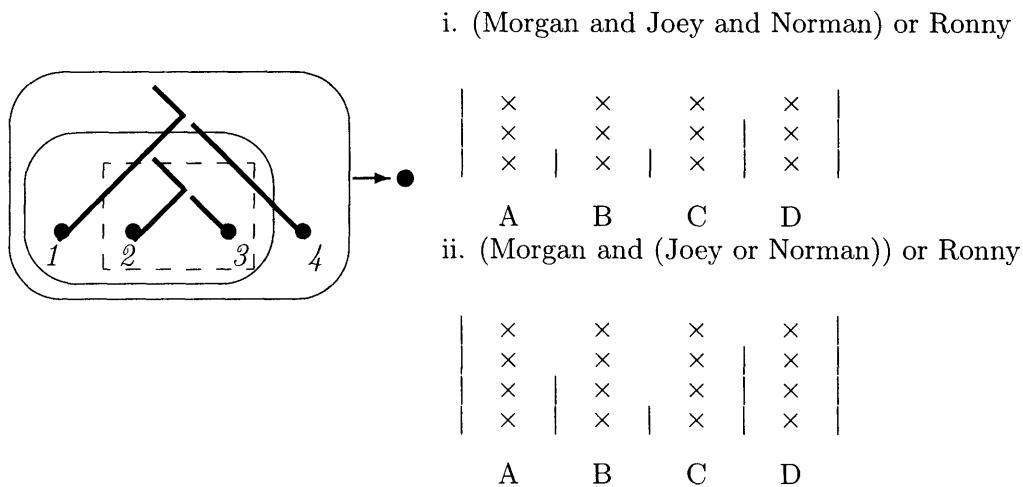
(94) Left-Branching: 1 Derivation



There are three binary branching tree involving 4 elements that have 1 right-branching node and one left-branching node. The elements on the right branch of a right-branching node can be either a separate cycle or be assembled together with the other material in the tree—which one is the right output depends on whether or not the associative law holds; the elements under the left branch on a left-branching node necessarily form a cycle together.

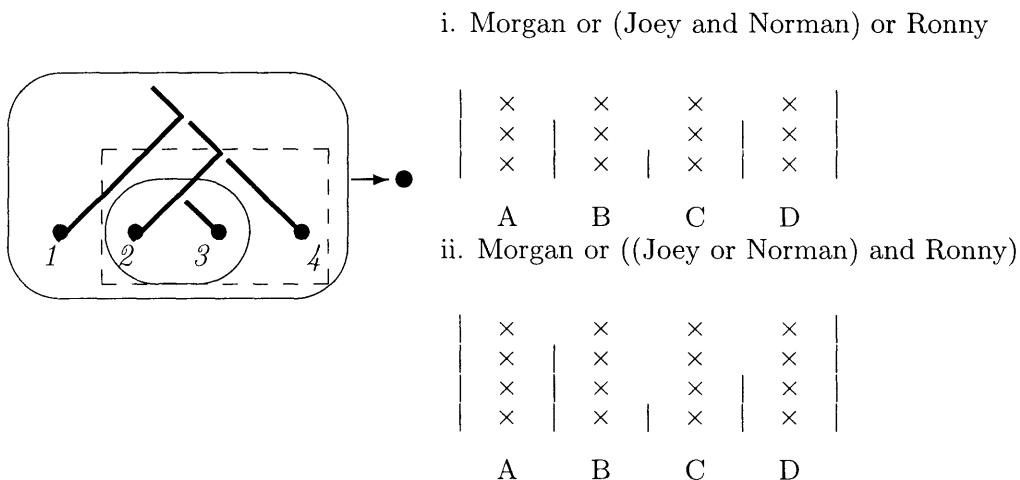
In the tree below, it is the first three elements that are on the left branch, and thus always form a separate cycle. Elements 2 and 3 may or may not form an additional separate cycle (a node that could be a separate cycle or be part of a bigger cycle, i.e. a right-branching node, is marked by the a dotted rectangle, which reflects the fact there could be an extra cycle—depending on the content of the terminal nodes and whether or not the associative law holds within the cycle that contains it):

(95) One Right-Branching node: 2 Derivations



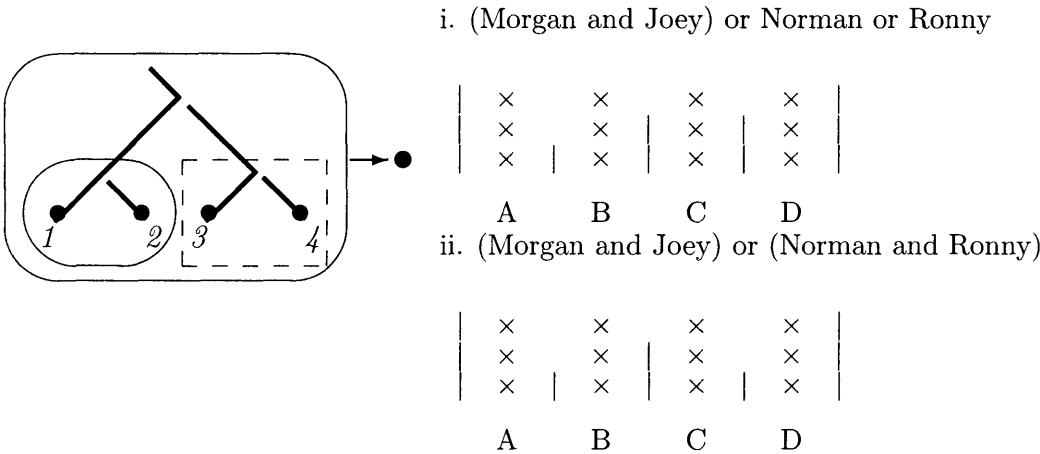
In the second structure, the middle two elements are on the left branch of the left-branching node and thus form a cycle. The last three structures may or may not form another separate cycle:

(96) One Right-Branching node: 2 Derivations



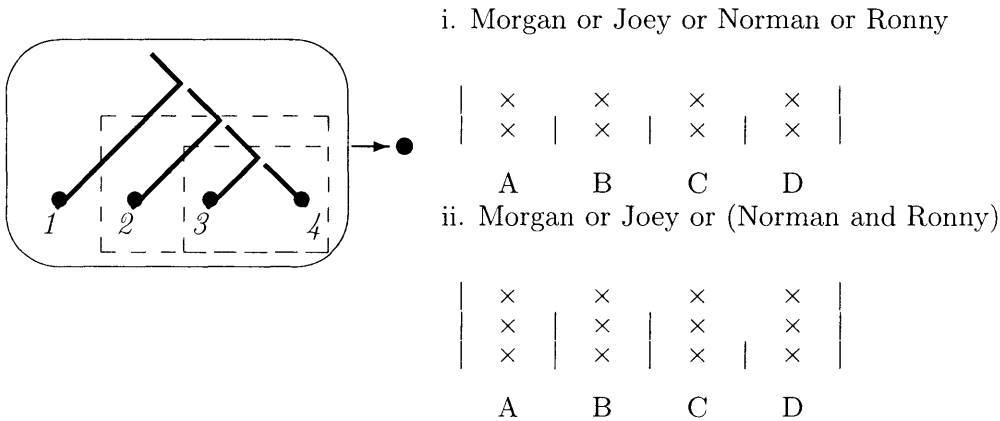
In the following tree structure, the first two elements are on the left branch of a left-branching node, and the last two elements can form an additional cycle:

(97) One Right-Branching node: 2 Derivations



Finally, the fully right-branching tree corresponds to 4 derivations, since each right branching node may or may not contain a separate cycle.

(98) Two right-branching nodes, $2 \times 2 = 4$ derivations



iii. Morgan or (Joey and Norman and Ronny)

	x		x		x		x	
	x		x		x		x	
	x		x		x		x	

A B C D

iv. Morgan or (Joey and (Norman or Ronny))

	x		x		x		x	
	x		x		x		x	
	x		x		x		x	
	x		x		x		x	

A B C D

The cyclic approach derives representations that reflect speaker's intuitions about grouping in coordinate structures. There are 11 different derivations, and correspondingly there are 11 different grids. The only mapping principle necessary is the one that the domain of a spell-out is mapped to a new foot in the grid (16). The grid comes about as a result of the way syntactic derivations interact with this mapping principle. The metrical representation reflects not just the tree structure, but also how it was assembled.

Summary

In the fragment of coordinate structures discussed here, prosodic phrasing can be derived from the way syntactic derivations work by the simple assumption that the output of cycles gets mapped to a foot in prosodic representations. Prosodic boundary ranks reflect syntax in a direct fashion. There is a one-to-one mapping between syntactic derivations and grid representations. Instead of viewing the grid as a separate representation, completely independent of syntax, we can see it as another way of representing syntactic information.

This is as expected according to the program for a metrical theory originally outlined in Liberman (1975, 258): "Thus the most basic assumptions of our theory depend on the idea that the phonological component is not so much a destruction of structure, which maps a complex tree onto a simple serial ordering of segments, as a transmutation of structure, which maps a structure suitable for operations in one

domain (syntax and semantics) onto a structure suitable for use in another domain (the motor control of articulatory gestures, and its perceptual analogue).”

While it is useful to think of the complete grid as an output of the grammar in order to state phonological generalizations such as phrasing and accent placement, it is not obvious whether or not the complete grid of a derivation is ever represented at a single point in time in the mind of a speaker. It is conceivable that less information is kept after spell-out, e.g. it might be that the only information kept is the relative rank of the current foot boundary relative to the starting point.

The presentation here assumes the minimalist framework, the presented recursive theory could be implemented in different way. For example, the findings lend themselves easily to be implemented in tree-adjoining grammar. The recursive steps necessary for this limited data set could be taken to be substitution operation, and the restriction of right-branching could be implemented as a restriction elementary trees in TAG. I will not explore the compatibility/incompatibility of the findings with alternative theories of syntax.

The discussion was entirely based on evidence from the domain of coordinate structure. Of course, it might be that the generalizations about carefully articulated coordinate structures are a special case and do not tell us anything about the interaction between syntax and prosody more generally. In chapter 3 argues, however, evidence is presented that the analysis extends to other domains the grammar.

The proposal in this chapter 2 captures some aspects of the prosody of coordinate structures, namely those that relate to relative boundary ranks. Consider the following constraint on the prosody of coordinate structures proposed in Taglicht (1998, 192):

(99) *Coordination Constraint (CC)*

For any coordinating node X, if any two daughter nodes of X are separated by an IP boundary, all the daughter nodes of X must be separated by IP boundaries.

This constraint is captured by the metrical representation derived here: it cannot be

the case that some boundaries between conjuncts at a given level of embedding in a coordination structure are intonational phrases and others are not, since this would entail that they are not separated by boundaries of equal rank. The prediction made here is even stronger: the boundaries separating the conjuncts must always be the of equal rank.

A question left open is how exactly the impressionistic boundary ranks are implemented phonologically and reflected in the phonetics of coordinate structures. For example, the lines in the metrical grid derived here are not labeled with respect to the intonational categories (intonational phrase, phonological phrase, etc.) which are standardly assumed in prosodic phonology (Selkirk, 1978, 1980, 1981, Nespor and Vogel, 1982, 1986). The representation derived here is an *unlabeled* grid, following the proposals in Liberman (1975), Selkirk (1984), Halle and Vergnaud (1987) and other works.

The theory does not make any predictions about which types of boundaries are actually used. In chapter 4 phonetic and phonological evidence is reviewed and an argument is presented that it is a desirable result that the syntax—phonology mapping only derives a relational prosodification, and does not provide prosodic labels.

The proposed generalization about coordinate structures is recursive in nature, and invokes *relative* boundary ranks. The mapping of the relative boundary ranks to phonological categories is a matter of implementation, and depending on speech rate and the place of a phrase marker in bigger syntactic context there is substantial variation as to how exactly the relative ranks are realized.

The alternative to this recursive and relational view would be a theory of syntax-to-phonology mapping that ties certain prosodic boundaries to certain prosodic categories (such as the intonational phrase). However, the data discussed in this paper involved recursion of the *same* syntactic category—Determiner Phrases. In other words, if the different ranks are realized with different prosodic categories, hence the source of these prosodic categories cannot be differences in the syntactic category of the objects that are coordinated.³⁶

³⁶Barry Schein points out that it would be conceivable to propose that the different prosodic

A theory that relates prosodic domains of a certain category with certain syntactic categories runs into problems in face of the data pattern observed in this chapter. A common way of implementing the idea of designated syntactic categories for prosodic domains is edge-marking or edge-alignment (Selkirk, 1986, Chen, 1987, Hale and Selkirk, 1987, Selkirk, 1995b, Truckenbrodt, 1995, 1999). Here an instantiation from Selkirk (1996, 444):

(100) The edge-based theory of the syntax-prosody interface

Right/Left edge of $\alpha \rightarrow$ edge of β ,

breaks in coordinate structures reflect the size of elided material, instead of being a reflex of recursive nesting. The idea is that in (1b), the first conjunct—‘Lysander’—is really an entire sentence ‘Lysander is in the forest’, or at least some bigger structure compared to (1a), while the second conjunct consists of two smaller conjuncts that only contain the material of the two DPs (and their event structure in Schein’s theory of event semantics):

- (1) a. Lysander | and Demetrius | and Hermia went to the forest.
- b. Lysander || and Demetrius | and Hermia went to the forest.

The strength of a boundary is correlated then with the size of the elided material. In this theory, it would indeed be possible to relate the strength of boundaries to differences in syntactic category (e.g. DP-coordination vs. CP-coordination).

It is entirely possible that some strong boundaries are a reflex of more elided structure. But it is not clear that this alternative hypothesis can explain all the data reported in this chapter. For example, it seems that even a strong boundary can be used in cases where only DPs are being coordinated (1a) should not have allow for a collective reading just like (2b):

- (2) a. Lysander || and Demetrius | or Hermia gathered in the forest.
- b. *Lysander gathered in the forest and Demetrius and Hermia gathered in the forest.

At least it is clear from the contrast with (2b) that the elided constituent is an entire IP. It is also not clear how such an alternative could cover the entire range of data regarding multiple levels of embedding, since it is unclear how to make the necessary distinctions when there is an additional level of embedding:

- (3) Lysander ||| and (Demetrius || (or Hermia and Helena).

α is a syntactic category, β is a prosodic category.

The edge of certain syntactic types of constituents, e.g. XP , are mapped to the edge of a certain type of prosodic constituent, e.g. phonological phrases (Φ).

Each conjunct in a coordinate structure involving names should be mapped to the prosodic category corresponding to XP s (maybe by XP -alignment). Say the designated category is the phonological phrase, Φ . But this would predict that the phrasing is always flat:

(101) * (Lysander) $_{\Phi}$ (and Hermia) $_{\Phi}$ (or Demetrius.) $_{\Phi}$

The hierarchical prosody that is actually observed constitutes ‘ XP -within- XP ’ effects. This point will be discussed in more detail in chapter 4. The relational view of syntax-to-prosody mapping is further confirmed by experimental evidence. The experiment and some evidence bearing on how the relational grid is implemented in on-line speech production is presented in chapter 5.

2.5 Appendix

2.5.1 Binary or Infinitary Connectors?

In our binary notation, we used the two labels XP vs. $\bar{X}$ to distinguish nodes that form their own cycle from nodes that are part of a bigger cycle. The proposed generalization was that nodes that are combined in such a way that the associative law holds can be computed in a single cycle.

But there is an alternative to this approach, which assigns the labels XP vs. $\bar{X}$ in a different way. We could instead treat the XP -nodes as projections of heads, which define their own cyclic domain. This has been proposed for ‘flat’ coordination in English, which were argued to constitute $\&Ps$ (also called Boolean Phrases or ConjPs), headed by a Boolean head (Munn, 1987, Collins, 1988, Zoerner, 1995). In the following, I will give evidence against this alternative.³⁷

³⁷An updated version of this account could be to posit $\&Ps$ and stipulate that they are spell-out

An observation that is often taken (cf. Progovac, 2003) to speak in favor of the &P approach is that in cases of iterated coordinated and disjunction, not all coordinators have to be pronounced:

- (102) a. Lysander, Demetrius, and Hermia.
 b. Lysander, Demetrius, or Hermia.

This occurrence of only a single ‘and’/‘or’ is unsurprising under the &P-approach, since there is only one head involved in the first place.

Of course, why the functor surfaces where it does is not obvious under this approach. Also, every connector in a flat structure *can* be pronounced, and there is no reason to assume that the two types of constructions differ radically in phrase structure or complexity. E.g., Frazier et al. (2000, 360) report results from an experiment comparing coordinate structures with elided and unelided connectors, and found no difference in reading times. Under the &P-story, the instances of ‘and’/‘or’ in addition to the final one would have to be analyzed as some kind of coordinate concord.³⁸

Another fact that could be taken to point to an &P analysis is that adverbials cannot take scope over two conjuncts unless they are linked by an overt connector. The argument would be that sentential adverbs cannot be placed inside of &Ps, just as they cannot usually be placed inside of some other XPs such as DPs.

- (103) a. Maybe Lysander, probably Demetrius, and definitely Hermia.
 b. * Maybe (Lysander Demetrius), and definitely Hermia.

Before looking more closely at the premise of those arguments, it should be noted that these data are not necessarily in conflict with the binary approach. Faced with (102), the approach that takes all connectors to be binary can invoke some notion of domains or ‘phases’. We would then have to make some stipulation that coordinate structures with arbitrarily many arguments are ‘spelled-out’ as a unit, maybe by stipulating some higher head which constitutes a ‘phase’ and triggers the spell-out of its complement.

³⁸According to Zoerner (1995, 27), the optional pronunciation of the connectors may be used to indicate some degree of emphasis.

phonological deletion or backwards ‘gapping’ of the connectors. It is not inconceivable that this alternative could be made to work.

The observation in (103), for example, can then be explained by requiring that in order for gapping of ‘and’/‘or’, the connectors *have to follow boundaries of equal rank*. We already observed above that adverbials induce prosodic boundaries around the conjuncts that they take scope over, so the connectors in (103) could not be gapped for prosodic reasons under this approach:

(104) Maybe (Lysander | and Demetrius), || and definitely Hermia.

The boundary preceding the last ‘and’ is stronger than the boundary preceding the first ‘and’. The arguments for the &P-approach over the binary approach are thus not conclusive.

The premise of the argument based on omitted connectors is that all the elements combined under a single &P projection are united by the same functional head, be it ‘and’ or ‘or’.

However, it is not clear that this premise is correct. The next section argues that it is incorrect. The section after that proposes an alternative account for structures with ‘omitted’ connectors, involving an unpronounced set-union operator ‘*’, which voids the need for infinitary functors.

Omitted Connectors are Different Connectors

The analysis within the &P approach requires conjunctive heads that can take any number of arguments. Suspiciously, coordination and conjunction seem to be the only functors that allow infinitary arguments.

Predicates in general do not allow this, e.g., there is no verb in English that has this property. An example would be a version of ‘precede’ or ‘is-an-ancestor-of’ that would take an arbitrary number of arguments and denote a completely ordered chain or a completely ordered line of ancestors. The following sentence means that A, B, and C each precede D, but not that the order is necessarily $A \prec B \prec C \prec D$:

(105) A, B, C precede D.

There are of course ways of expressing a complete ordering, but they involve coordination, and in order to obtain a complete order chain of precedence meta-linguistic reference to linear order with the adverbial ‘in that order’ is necessary:

(106) A, B, C and D, in that order, precede one another.

It seems that (105) involves a coordination structure without any overt coordinator at all, and maybe it does not sound quite complete without any overt ‘and’. But certainly the zero connector involved is not the predicate ‘precede’.

There is more evidence that there are zero connectors that figure into a structure similar to (102), but are clearly not derived by combining n arguments by the same operation.

The functor ‘through’ is an example. ‘Through’ cannot be used with items that are adjacent on a list (107a), and yet (107b) is grammatical (Assuming that in the context the relevant domain is the alphabet, and A directly precedes B). The zero-functor involved cannot be ‘through’ itself.

- (107) a. * A through B.
b. A, B, C through Z.

If it is not the functor ‘through’ that connects the other arguments, there must be some covert connector that does. Another similar case is ‘but’, which is used instead of ‘and’ in contexts where the two adjacent conjuncts differ in polarity (Horn, 1996).

(108) Who went to the forest?

Lysander, Demetrius, Helena, but not Hermia.

Also, there are structures that show zero connectors without any overt functor at all, called asyndetic coordination (as opposed to syndetic coordination):³⁹

- (109) With bracelets of thy hair, rings, gauds, conceits,
knacks, trifles, nosegays, sweetmeats

³⁹Barry Schein points out that this asyndetic coordination could be the deliberate rhetorical use of an incomplete, interrupted coordination. This could apply to all cases of asyndetic coordination in English. I am not sure how to rule out this possibility.

—messengers of strong prevailment in unhardened youth.

(A Midsummer Night's Dream, I.1 36–38)

Another argument is related to prosodically conditioned allomorphy in coordinate structures. Zoerner (1995, 28) observes that if all coordinators are realized, they have to be identical in form. The judgements presuppose a flat prosody:

- (110) a. Robin | and Kim | and Terry.
b. Robin'n | Tim'n | Terry.
c. * Robin | and Kim'n | Terry.
d. * Robin'n | Kim and | Terry.

The prosodic allomorph of 'and' is used to encode the strength of the following boundary. Since in 'flat' structures the conjuncts are separated by boundaries of equal rank, the allomorphs have to be identical. Relating the allomorph of the connectors to the strength of the following boundary can be further motivated by the fact that it is the lowest level of embedding that takes the prosodically weaker allomorph, and not vice-versa:

- (111) a. ?? Robin and | Kim'n || Terry.
b. ?? Robin'n || Kim | and Terry.

The allomorphs of 'and' are thus used in a way that reflects the grammatically assigned boundary rank. The only exception to this generalization about the distribution of allomorphs of 'and'—phonological identity at boundaries of equal rank—would be the case were the last functor is realized and all the others remain unpronounced.

Finally, coordinate structure with overt connectors are not identical in their grammatical properties to those with omitted connectors. To illustrate this, an excursion into NPI-licensing is necessary.

NPI-Licensing

Consider the following example:⁴⁰

⁴⁰Progovac (2003) erroneously claims that NPI-licensing is not possible in coordinate structures. One example used as evidence involves the conjunction 'and', which is an intervener for NPI-licensing,

- (112) Did they find anything suspicious?
- a. No weapons or any drugs or any other sign of a crime.
 - b. * Any weapons, or no drugs, or any other sign of a crime.

NPIs are licensed from the first disjunct into later disjuncts, but not vice-versa, as expected under the right-branching account for prosodically flat coordinate structures outlined here.

There is an interesting contrast, however, depending on whether ‘or’ is pronounced: While a negative quantifier in the first disjunct can license NPIs in later

as discussed in the context of example (1).

- (1) * He chased nobody and any dogs.

The connector to use to test for NPI-licensing is ‘or’, which is not an intervener. Using ‘or’, however, does not improve the example much according to Progovac (2003)—but this example might be odd for a number of reasons. A confounding factor is that it is marked to use a negative quantifier in direct object position, instead of using sentential negation. A further problem might be that ‘nobody’ instead of ‘no human being’ is used to contrast with ‘dog’. NPI-licensing is fine in the following case:

- (2) No cat or any dog was chased.

There are also many naturally occurring examples (from Google, and then tested on native speakers) with NPI licensing in coordinate structures in direct object position:

- (3) a. I could get no help, or any response from the on-line customer service.
 b. ... and enclose no money or any checks at this time.

I started my own print magazine
 in this town and got no help or any replies from anybody in this town.

The following example illustrates that ‘nobody’ can also license NPIs (again a naturally occurring example found with Google, then tested on native speakers):

- (4) He warned that nobody or any amount of intimidation would stop the community from pursuing its cultural heritage.

disjuncts (113a), is impossible to repeat the negative quantifier instead of using an indefinite (113b).

- (113) a. No weapons, or any drugs, or any money was found at the site of the crime.
b. *No weapons, or no drugs, or no money was found at the site of the crime.

The facts dramatically change when we drop ‘or’. It turns out that NPI-licensing is only possible into a disjunct that is headed by an *overt* ‘or’:⁴¹

- (114) a. ?*No weapons, any drugs, or any money was found at the site of the crime.
b. No weapons, no drugs, or any money was found at the site of the crime.

The pattern observed between the first two conjuncts in (114a,b) is identical to that observed in the case of coordination:⁴²

- (115) a. *No weapons and any drugs were found at the site of the crime.
b. No weapons and no drugs were found at the site of the crime.

The explanation for the ungrammaticality of (115a) offered in Linebarger (1980) is that ‘and’ is an intervener and blocks NPI-licensing. Another way of looking at it is to say that ‘and’ is a PPI. If this is the correct explanation, (114) raises the following puzzle: if ‘or’ is not an intervener, why is the zero version of ‘or’?

⁴¹Or alternatively the negative quantifier can be altogether omitted—the use of ‘any’ is of course also not obligatory in the last disjunct either.

⁴²Schein (2001, 12) reports some examples in which an NPI can be licensed within a coordination:

- (1) No wolf mother and any of her cubs remain close for very long.

(1) is at least much better than (115). This seems to be related to the fact that the negative quantifier quantifies over all possible mother/cub pairs, rather than the set of the mother and all of the cubs.

In sum, when looking at conjunctions/disjunctions, it is doubtful whether the zero-connectors are really the equivalent of the overt connectors. Thus the premise for the argument for n-ary conjunction and disjunction is dubious at best.

A List Operator

Another way to look at the ‘dropped’ connectors is that the connector that relates the conjuncts in the absence of an overt ‘and’/‘or’ is *neither* of those two, but instead is a set-union or list operator ‘*’ which combines elements into a single set.⁴³ The operator ‘*’ connects two arguments by set union and is involved in the creation of lists:

$$(116) \quad [[*]] (A)(B) = \{A, B\}$$

We can decompose ‘and’ into two parts: a universal quantifier, and a set-union connector ‘*’; similarly, we can decompose ‘or’ into an existential quantifier, and the connector ‘*’. This was suggested to me by Marta Abrusan. For a decomposition of disjunction into an existential quantifier and a set union operation see Alonso-Ovalle (2005), who proposes that the set that disjunctions create can be bound by various different operators, in analogy to the case of indefinites in the approach of Heim (1982).

In order to get the right meaning for both ‘and’ and ‘or’, we have to be able to let different quantifiers take scope over the conjuncts:

- (117) a. A and B: $\forall x.x \in A * B$
 b. A or B: $\exists x.x \in A * B$

The connectors ‘and’ and ‘or’ are allomorphs of ‘*’. They they agree with a c-commanding quantifier. The precise conditions on when ‘and’, ‘or’ and $\emptyset$ and whatever other possible agreement forms there may be (e.g. ‘nor’) are intricate and I cannot discuss this issue at this point.

⁴³At least we follow Schwarzschild (1996) in allowing individuals to be represented either as such or as sets containing themselves we can treat ‘*’ as simple set union.

- (118) a. $A, B, \text{ and } C = (A * B * C): \forall x.x \in A * B * C$
 b. $A, B, \text{ or } C = (A * B * C): \exists x.x \in A * B * C$

The operator ‘*’ itself is associative and binary. It creates sets. These sets are given in the form of lists, with a right-branching structure and an iteration of binary ‘*’. Lists without overt connectors in general can combine with predicates. The meaning in those cases is just as what one would expect if they are combined by the list operator ‘*’.

The following expression states that each member of the list precedes ‘Z’ and expresses a partial ordering, but it does not denote a precedence chain with a complete ordering:⁴⁴

- (119) A, B, C, F, E, D precede Z.

The reason why (119) might in general be dispreferred compared to an alternative with an overt ‘and’ and ‘or’ might be that it leaves open how to quantify over the set created by the list:

- (120) bracelets of thy hair, rings, gauds, conceits, knacks, trifles, nosegays, sweet-meats.

The approach using ‘*’ voids the the necessity of functors with infinitely many arguments and voids the need to commit us to treat zero-connectors and and/or as identical, both of which are problematic assumptions.

The account using ‘*’ *explains* why there are only a few items that appear to be able to take any number of arguments: they are all pronunciations of ‘*’ that agree with different operators that can quantify over the set. The inventory of operators is a

⁴⁴‘etc.’ could be seen as a function that takes a single argument which be a list. The following would then be another instance of a list without overt connective:

- (1) A, B, C, etc.

Zoerner (1995) offers an alternative interpretation, and interprets ‘etc.’ as a pronoun standing in for ‘[and X]’.

closed set, and probably very small. Cross-linguistically, there is quite some variation in how connectors can be pronounced, but the inventory does not seem to be involve open class items (Haspelmath, 2004).

2.5.2 A Pragmatic Principle ‘Avoid Ambiguity’?

Sequences of 2+ elements in which the associative law does not hold are ambiguous. Inserting prosodic boundaries to signal the bracketing disambiguates these structures. Grice (2005) proposes the principle ‘Avoid Ambiguity’ as one of his maxims of manner. Is prosodic disambiguation due to a pragmatic principle?

The definition of level of embedding in terms of the associative law is formulated irrespective of whether or not in a particular utterance context there is any danger of an ambiguity. The reason is that under the pragmatic approach one might expect that sufficient contextual information would void the need to realize prosodic phrasing. This would mean that (38) is too weak, since it excludes the possibility of having a prosodically flat structure in cases of non-associative combinations, where information from the context disambiguates the structure. But evidence from the literature of prosodic disambiguation shows that disambiguating prosodic cues are placed *regardless of whether or not the context can be used to disambiguate a structure*.

In natural language it is less common that each conceivably possible bracketing is equally likely, as is often the case in logical formulas. Orthography illustrates that it is possible to get rid of prosodic information and still communicate—omitting disambiguating parentheses in logical formulas can easily make the content unrecoverable. In orthography, prosodic information is at least minimized and only sporadically encoded with punctuation.⁴⁵

⁴⁵In fact, according to (Nunberg, 1990), punctuation obeys its own textual grammar and cannot be conceived of as a deteriorated version of spoken prosody (certain correlations notwithstanding, as reflected by the term ‘comma-intonation’ often used in the context of the intonational realization of certain modifiers, parentheticals, and appositives (e.g. Potts (2005))). See Nunberg (1990, 39) for the use of punctuation to disambiguate coordination structures.

Orthography thus shows that prosodic structure *is* not necessary to be able to communicate, and that at least in writing we convey meaning without an explicit prosodic structure. The question is whether in speech, given a context that would disambiguate a structure, prosody is still produced by the speaker.

Several authors have suggested that prosody might only be used in cases where a speaker is aware of an ambiguity and intends to disambiguate a structure Lehiste (e.g. 1973, footnote 1). Some authors proposed an ‘audience design’ model of prosodic disambiguation, in which speakers use prosodic cues for disambiguation in order to help the addressee in decoding the message (see Krajic and Brennan, 2005, for a summary). Some experiments report that prosodic cues are only inserted when an ambiguity exists (Allbritton et al., 1996, Snedeker and Trueswell, 2003), which could speak in favor of a pragmatic account.

But experiments that create situations in which speakers actually interact with an addressee—in other words, experiments that try to create a natural communicative situation—show that prosodic cues are placed independently of whether or not context helps to disambiguate structure and irrespective of the needs of the addressee in the particular situation (Schafer et al., 2000, Krajic and Brennan, 2005).⁴⁶

In the experiment on coordinate structures reported in chapter 5, ample context was provided, but prosodic phrasing was realized nonetheless. It was not a very natural context, and the subjects knew their interlocutor did not have to understand the message based on their utterance.

Therefore, I will assume in the following that the generalization about prosody and associativity is a grammatical generalization that holds true independent of context. The prosodic boundaries discussed here are a result of the way grammar assembles structure, and a reflex of the syntactic derivation. A cycle is induced if the logical properties of the elements involved is such that the associative law does not hold—

⁴⁶Krajic and Brennan (2005) suggest that subjects in the experiment in Snedeker and Trueswell (2003) have fallen into ‘routine, uninformative prosody’. The results of the production experiment in chapter 5 suggest that prosodic structure can be ‘washed out’ in such a way that sub-distinctions between weaker boundaries are not realized.

whether or not a speaker is aware of an ambiguity.

The view that takes prosody to be shaped by syntax is taken in much of the literature on the phonetics of prosody (Klatt, 1975, Cooper and Paccia-Cooper, 1980, Gee and Grosjean, 1983, Ferreira, 1988), and also the standard assumption in the literature on phrasal phonology in generative grammar (Selkirk, 1984, Nespor and Vogel, 1986, Inkelas and Zec, 1990, Cinque, 1993, Truckenbrodt, 1995).